A FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

Abstract. A countable discrete group is MF if it embeds in the unitary group of a matrix quotient $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$; equivalently, if it admits finite-dimensional approximate unitary representations, asymptotically multiplicative in operator norm, that separate every nontrivial element from the identity. We construct an explicit finitely presented group E that is not MF, refuting the conjecture that every countable group is MF.

More generally, let Γ be a finitely presented property-(T) group with a proper injective endomorphism α. From (Γ, α) we build a finitely presented group with a nontrivial central involution w such that $\Theta(w) = 1$ for every homomorphism Θ to $\mathcal{U}(\mathcal{Q})$. In every operator-norm asymptotic representation, one-sided conjugation preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm. If $\Theta(w) \neq 1$, compressing to the (-1)-spectral projection of $\Theta(w)$ yields a contradiction. Taking $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and $\alpha(v, A) = (2v, A)$ gives the eight-generator, forty-one-relator group E.

The group E is sofic, hence hyperlinear, so neither soficity nor hyperlinearity implies the MF property. Its reduced C^*-algebra $C^*_{\mathrm{r}}(E)$ is separable, exact and stably finite but not MF. The quotient $E/\langle w \rangle$ is also non-MF, so $\{1, w\}$ is properly contained in the intersection of the kernels of all homomorphisms $E \to \mathcal{U}(\mathcal{Q})$.

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Contents

1. Introduction

The *MF conjecture* asks whether every countable discrete group admits finite-dimensional matrix models whose multiplicative errors tend to zero in operator norm [Sh]. No counterexample was previously known [Sc, Sh]. We construct an explicit eight-generator, forty-one-relator group that is not MF. Residually finite groups are MF because their finite quotients separate points; LEF groups are MF [CDE], and quasidiagonality implies that amenable groups are MF [TWW]. Recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP].

The result.

Theorem A (an explicit finitely presented non-MF group)**.** *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the forty-one relators displayed in Definition 6.1, and put*

$$w = [tct^{-1},\, v_1(tct^{-1})v_1^{-1}].$$

The element w is central and satisfies $w^2 = 1$.

1. *$w \neq 1$ in E, so w is a central involution;*
2. *for every sequence (d_n) of positive integers and every homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q}((d_n)))$ into the unitary group of the corresponding matrix quotient, one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C^*_{\max}(E)$ nor $C^*_{\mathrm{r}}(E)$ is an MF C^*-algebra.

Outline of the proof. All approximation errors are measured in operator norm. The normalized Hilbert–Schmidt norm is used for the Hilbert spaces on which the conjugation representations act.

Regard $M_d(\mathbb{C})$ as the Hilbert space $L^2(M_d(\mathbb{C}), \mathrm{tr}_d)$, on which conjugation by a unitary acts unitarily. For unitaries u_1, u_2, uniformly in d,

$$\|\mathrm{Ad}\, u_1 - \mathrm{Ad}\, u_2\|_{B(L^2(M_d(\mathbb{C}),\mathrm{tr}_d))} \;\leq\; 2\,\|u_1 - u_2\|,$$

so conjugation turns an operator-norm asymptotic representation into an operator-norm asymptotic representation on the Hilbert–Schmidt spaces.

Let Γ have property (T), let $\iota\colon \Gamma \to H$ be a homomorphism, and let $t, c \in H$ satisfy

$$t\,\iota(\Gamma)\,t^{-1} \subseteq \iota(\Gamma), \qquad [c, \iota(\Gamma)] = 1.$$

In a norm ultraproduct B_ω of the blocks, the classes $\pi(g) = [\mathrm{Ad}\, U_n(g)]_\omega$ define a unitary representation of H. Property (T) gives a spectral gap for the Hermitian average h of the generators on the orthogonal complement of the $\iota(\Gamma)$-invariant vectors. Thus the spectral projection $P = \chi_{\{1\}}(h)$ belongs to B_ω and projects onto the invariant subspace. Writing $V = \pi(t)$, the inclusion above implies $P \leq VPV^*$; the two projections are Murray–von Neumann equivalent, and B_ω is finite, so $P = VPV^*$. Hence conjugation by t preserves the asymptotic commutant of $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm (Theorem 3.4).

Put $d = tct^{-1}$ and $u = [\,d, \iota(a)\,]$ for $a \in \Gamma$. Since c commutes with $\iota(\Gamma)$, the sequence $(U_n(c))$ lies in that asymptotic commutant. Theorem 3.4 therefore gives $\|[U_n(d), U_n(\iota(\gamma))]_-\|_2 \to 0$ for every $\gamma \in \Gamma$. Consequently, in every operator-norm asymptotic representation (U_n) of H,

$$\|U_n(u) - 1\|_2 \longrightarrow 0.$$

Suppose now that $w = u^2$ is central in H with $w^2 = 1$, and let Θ be a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with $\Theta(H)$. After choosing unitary lifts, functional calculus gives projection lifts $q_n \in M_{d_n}(\mathbb{C})$ that are nonzero for infinitely many n. Compressing to these corners and using their normalized traces gives an operator-norm asymptotic representation of H in which the image of w converges to -1 in operator norm. There the display above gives $\left\|U_n(u)^2 - 1\right\|_2 \to 0$, while $\|-1-1\|_2 = 2$. Hence $\Theta(w) = 1$, and if $w \neq 1$ then H is not MF (Theorem 4.1).

Let Γ be a finitely presented Kazhdan group with a proper injective endomorphism α, let $a \in \Gamma \setminus \alpha(\Gamma)$, and adjoin t and c subject to $t\gamma t^{-1} = \alpha(\gamma)$, $c^2 = 1$ and $[c, \gamma] = 1$. In the direct limit of α the cosets $t\Gamma$ and $at\Gamma$ are distinct, since $t^{-1}at \in \Gamma$ would give $a \in \alpha(\Gamma)$. Let the ambient group act on the Clifford group by permuting the generators indexed by the coset space, and map c to the generator at the base coset. Then d and ada^{-1} map to the generators indexed by the two distinct cosets. Distinct Clifford generators anticommute, so u maps to a product of two anticommuting involutions and $w = u^2$ to -1 (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I)$$

gives the group E of Theorem A.

The same one-sided inclusion also yields exact finite-dimensional statements. For a homomorphism π to a finite group, $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$, since the left side is a conjugate of the right side contained in it and the two are finite of the same cardinality; so $\pi(tct^{-1})$ centralizes $\pi(\Gamma)$ and every commutator $[\,tct^{-1}, a\,]$ with $a \in \Gamma$ lies in $\ker \pi$. For a finite-dimensional representation over any field the same conclusion holds with the commutant C of $\pi(\Gamma)$ in place of the subgroup: the inclusion gives $C \subseteq \pi(t)\, C\, \pi(t)^{-1}$ between subspaces of equal finite dimension, hence equality, and $\pi(c) \in C$ then puts $\pi(tct^{-1})$ in C (Theorem 3.3). In the norm-ultraproduct argument, finiteness gives the corresponding conclusion: the equivalent projections $P \leq VPV^*$ are equal.

The argument uses property (T) for the subgroup $\iota(\Gamma)$. The group E itself maps onto $\mathbb{Z}$ by sending t to 1 and every other generator to 0, and therefore does not have property (T).

The reduced group C^*-algebra $C^*_{\mathrm{r}}(E)$ is separable and stably finite but not MF (Theorem C); it is also exact. The group E is sofic (Theorem D), so neither soficity nor hyperlinearity implies the MF property. The quotient

$E/\langle w\rangle$ is not MF either, so $\{1, w\}$ is a proper subgroup of the MF residual of E (Theorem B).

Relation to prior work. The MF problem for C^*-algebras predates the group question. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable C^*-algebra was known not to be MF [BK, CDE]. Their natural candidate was $C^*_r(F_2)$, which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently implied the existence of abstract stably finite non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group C^*-algebra fails to be MF [Sh].

In his 2018 ICM address, Thom asked for which Schatten p-norms there exist non-approximable groups [Th, Question 3.11]. The case $p = 2$ was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases $1 < p < \infty$ by Lubotzky–Oppenheim [LuO], and the case $p = 1$ by Bachner–Dogon–Lubotzky [BDL]. Those results left three approximation problems open: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), or MF (unitaries in operator norm). The first was answered negatively in 2026: a nonsofic group was constructed in [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. The present paper answers the third negatively; at the time of writing the hyperlinear problem is open. Our construction likewise uses a property-(T) subgroup Γ with $t\Gamma t^{-1} \subsetneq \Gamma$. In the operator-norm setting, conjugation by t preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm. Throughout, "MF" is the operator-norm property of Carrión–Dadarlat–Eckhardt [CDE], in the form used by Shulman and by Bachner–Dogon–Lubotzky [Sh, BDL]; Section 2 fixes it and its equivalent formulations.

A closely related setting is the recent work on nonsofic groups. One-sided conjugation of a property-(T) subgroup, together with an element centralizing that subgroup, appears in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. Here the group acts by conjugation on the Hilbert–Schmidt space of each matrix block, and finiteness of the norm ultraproduct as a C^*-algebra forces the projection onto the $\iota(\Gamma)$-fixed subspace to equal its unitary conjugate (Theorem 3.4).

Slofstra's hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3]. He also considers the elements sent to the identity by every exact finite-dimensional representation and those sent to the identity by every approximate representation [Slo17, Definitions 2.5–2.7]; the two subgroups they form are the exact and the approximate analogues of the MF residual used below. Negative-eigenspace compression appears in

Slofstra–Vidick [SV, proof of Proposition 2.7], and Fritz's marked $\mathrm{BS}(2,3)$ example is the context for the cyclic comparison of Section S7 [Fri, SV]. Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedures used in Section S5 [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5], and handle a finite normal subgroup by a character-isotypic corner [BDL, proof of Proposition 1.6]; Theorem S5.3 uses that corner together with one-sided conjugation of a property-(T) subgroup.

Bekka and Bekka–Valette study property (T) through the conjugation representation $\pi \otimes \bar{\pi}$ on Hilbert–Schmidt space [Be, BV], and Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels' group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

MF for a group C^*-algebra implies MF for the underlying group, but the converse implication need not hold. For the algebra one asks for an embedding $C^*_{\mathrm{r}}(E) \hookrightarrow \mathcal{Q}$ of C^*-algebras; for the group one asks only for an injective homomorphism $E \to \mathcal{U}(\mathcal{Q})$, which need not extend to $C^*_{\mathrm{r}}(E)$. Failure of MF for a group C^*-algebra therefore leaves open whether the group itself embeds. For E, Theorem A rules out both.

2. Matrix quotients

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. For group elements $[g,h] = ghg^{-1}h^{-1}$; for operators $[x,y]_- = xy - yx$. A C^*-homomorphism is unital only when this is stated; in particular an MF embedding may be nonunital. For a unital C^*-algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r, the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$. For a nonzero projection $q \in M_r(\mathbb{C})$ we equip the corner $qM_r(\mathbb{C})q$ with its own normalized trace. When the trace must be specified, we write $\|x\|_{2,\mathrm{tr}_r}$. The unnormalized trace is $\mathrm{Tr} = r \cdot \mathrm{tr}_r$ on $M_r(\mathbb{C})$. We repeatedly use the inequalities $\|x\|_2 \le \|x\|$; $|\,\mathrm{tr}_r(x)| \le \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v.

Given a sequence $(d_n)_{n\in\mathbb{N}}$ of positive integers, write

$$\mathcal{Q}((d_n)_{n\in\mathbb{N}}) \;=\; \prod_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}) \Big/ \bigoplus_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}), \tag{1}$$

where the product is the bounded C^*-product and $\bigoplus_n$ is the c_0-sum, the ideal of sequences with $\|x_n\| \to 0$; equivalently, $\mathcal{Q}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$. Write $q\colon \prod_n M_{d_n}(\mathbb{C}) \to \mathcal{Q}$ for the quotient map. Following Blackadar–Kirchberg [BK], a separable C^*-algebra is *MF* if it embeds into an algebra of the form (1).

The following lifting lemmas are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 2.1 (lifting)**.** *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

Proof. Lift u to a bounded sequence (x_n); unitarity of $q((x_n))$ gives $\|x_n^* x_n - 1\| \to 0$. Once $\|x_n^* x_n - 1\| \le \frac{1}{2}$, the matrix x_n is invertible and the polar correction $u_n = x_n(x_n^* x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \le 2\,\|x_n\|\,\|x_n^* x_n - 1\| \to 0$ by continuous functional calculus; set $u_n = 1$ at the finitely many remaining indices. ☐

Write

$$\mathcal{N} = \{(u_n) \in \textstyle\prod_n \mathcal{U}(d_n) \;:\; \|u_n - 1\| \longrightarrow 0\}, \tag{2}$$

a normal subgroup of $\prod_n \mathcal{U}(d_n)$.

Lemma 2.2 (unitaries in the quotient)**.** *For every sequence (d_n), with $\mathcal{N}$ as in (2), the map*

$$\kappa_{(d_n)} \colon \prod_n \mathcal{U}(d_n)/\mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \qquad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. Thus a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ may be represented by unitary lifts $U_n(g)$ whose multiplicative defects tend to 0 in operator norm, with two lift sequences identified when their quotient tends to 1.

Proof. The kernel of $(u_n) \mapsto q((u_n))$ is exactly $\mathcal{N}$, so the induced map is injective. Surjectivity is Lemma 2.1. ☐

An *operator-norm asymptotic representation* of H is a sequence of maps $U_n \colon H \to \mathcal{U}(d_n)$ such that $\|U_n(g)U_n(h) - U_n(gh)\| \to 0$ for all $g, h \in H$. Taking $g = h = 1$ gives $\|U_n(1) - 1\| \to 0$ automatically, since $U_n(1)$ is unitary. By Lemma 2.2, choosing unitary lifts $U_n(g)$ of $\Theta(g)$ turns a homomorphism $\Theta \colon H \to \mathcal{U}(\mathcal{Q})$ into such a sequence, and every such sequence arises this way. A countable group is *MF* if it admits an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ for some sequence of dimensions.

Proposition 2.3 (Equivalent formulations of the MF property)**.** *For every countable group, the group-MF definition above — an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ — is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

The notion of approximate unitary representation used in Proposition 2.3 is defined in Section S3. Two stronger conventions also occur in the literature: in [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring element, and the purely matricial field (PMF) property of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

3. One-sided conjugation in matrix models

In finite dimensions, the commutant C of the base satisfies $C \subseteq \pi(t)C\pi(t)^{-1}$; equality follows because the two spaces have the same finite dimension. In the norm-ultraproduct argument, the fixed-space projection satisfies $P \leq VPV^*$ by $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$; the two projections are Murray–von Neumann equivalent, and finiteness of the ultraproduct gives $P = VPV^*$.

Lemma 3.1 (comparison in a finite algebra)**.** *Let A be a finite unital C^*-algebra and let $p, q \in A$ be projections with $p \leq q$ and $p \sim q$. Then $p = q$.*

Proof. Let $r \in A$ satisfy $r^*r = q$ and $rr^* = p$. From $r^*r = q$ we get $r = rq$. Since $p \leq q$ we have $qp = p$, so

$$pr = rr^*r = rq = r, \qquad qr = q(pr) = (qp)r = pr = r,$$

and $r = qr$ as well. Hence the cross terms in $\sigma^*\sigma$ vanish for $\sigma = r + (1-q)$, and $\sigma^*\sigma = q + (1-q) = 1$, so σ is an isometry. Finiteness makes σ unitary, and $\sigma\sigma^* = p + (1-q) = 1$ gives $p = q$. ☐

Lemma 3.2 (finiteness of a norm quotient of matrix blocks)**.** *Let (K_n) be finite-dimensional Hilbert spaces and let*

$$B_{\mathrm{c}} = \prod_n B(K_n) \Big/ \bigoplus_n B(K_n)$$

be the quotient of the bounded family algebra by the c_0-sum. Then B_{c} is a finite C^*-algebra: every $\sigma \in B_{\mathrm{c}}$ with $\sigma^*\sigma = 1$ satisfies $\sigma\sigma^* = 1$.

Proof. Lift σ to a bounded family (σ_n). Then $\|\sigma_n^*\sigma_n - 1\| \to 0$, so for all large n the Gram defect is at most $\frac{1}{2}$ and $\sigma_n^*\sigma_n$ is invertible; each σ_n acts on a finite-dimensional space, so polar correction replaces it by a unitary $w_n = \sigma_n(\sigma_n^*\sigma_n)^{-1/2}$ with $\|\sigma_n - w_n\| \to 0$. Hence σ is the class of (w_n), and $w_n w_n^* = 1$ at every such n gives $\sigma\sigma^* = 1$. ☐

Let ω be a free ultrafilter on $\mathbb{N}$, let K_ω be the Hilbert-space ultraproduct of the K_n, and let $B_\omega = \prod_\omega B(K_n)$ be the norm ultraproduct, acting on K_ω by $[A_n]_\omega[\xi_n]_\omega = [A_n\xi_n]_\omega$. The same polar-correction argument shows that B_ω is finite, with ordinary eventual convergence replaced by convergence along ω. Its action on K_ω is faithful: if $A = [A_n]_\omega \neq 0$ then $\lim_\omega \|A_n\| = \delta > 0$, and unit vectors ξ_n with $\|A_n\xi_n\| \geq \|A_n\| - 1/n$ give $\|A[\xi_n]_\omega\| = \delta$. Hence for projections $P, Q \in B_\omega$ the inclusion $\operatorname{ran} P \subseteq \operatorname{ran} Q$ makes QP and P act identically on K_ω, so $QP = P$, that is, $P \leq Q$; the converse is immediate.

The exact case.

Theorem 3.3 (the finite-dimensional case)**.** *Let k be a field, V a finite-dimensional k-vector space, H a group, $\Gamma \leq H$ a subgroup, $t, c \in H$, and $a \in \Gamma$. Assume that $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi\colon H \to \mathrm{GL}(V)$ satisfies*

$$\pi\big(\big[\,tct^{-1},\ a(tct^{-1})a^{-1}\big]\big) \;=\; 1.$$

More generally, every such π is trivial on the subgroup $\mathfrak{D}(H,\Gamma)$ of (9).

Proof. Let

$$C = \{x \in \mathrm{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\mathrm{End}_k(V)$, and write $\Phi(x) = \pi(t)\, x\, \pi(t)^{-1}$ for conjugation by $\pi(t)$, a linear automorphism of $\mathrm{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and put $y = \pi(t)^{-1} x\, \pi(t)$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x, and therefore

$$\pi(\gamma)\, y = \pi(t)^{-1}\, \pi(t\gamma t^{-1})\, x\, \pi(t) = \pi(t)^{-1}\, x\, \pi(t\gamma t^{-1})\, \pi(t) = y\, \pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$C \;=\; \Phi(C) \;=\; \pi(t)\, C\, \pi(t)^{-1}. \tag{3}$$

Now $\pi(c) \in C$ because c centralizes Γ, so $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$ by (3). Since $a \in \Gamma$, the operator $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi\big(a(tct^{-1})a^{-1}\big) = \pi(tct^{-1})$ and

$$\pi\big([\, tct^{-1},\ a(tct^{-1})a^{-1}]\big) = [\, \pi(tct^{-1}),\ \pi(tct^{-1})] = 1.$$

The subgroup-valued assertion follows by applying the same commutant argument to every one-sided conjugator; the details are given in Section S6. ∎

Every finite-dimensional linear representation of E over any field maps w to the identity; consequently every homomorphism from E to a finite group does as well (Corollary S6.1).

In an MF model, multiplicativity is only asymptotic, so the coordinate fixed spaces need not be invariant. Operator-norm control makes the adjoint actions asymptotically multiplicative on the Hilbert–Schmidt spaces. The resulting Kazhdan projection lies in a finite norm ultraproduct, where $P \leq VPV^*$ and $P \sim VPV^*$ imply $P = VPV^*$. The property (T) hypothesis is essential: the group E_{BS} of Section S7 has cyclic base, its distinguished word is trivial in every finite-dimensional representation over every field, and some homomorphism $E_{\mathrm{BS}} \to \mathcal{U}(\mathcal{Q})$ maps that word to a nontrivial element.

Theorem 3.4 (one-sided conjugation preserves asymptotic commutants)**.** *Let Γ and H be groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, let $s \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

Let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and asymptotically centralizes $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm,

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma))x_n\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma),$$

then the conjugated sequence does as well:

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]_-\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma).$$

Proof. Suppose the conclusion fails: there are $\gamma_0 \in \Gamma$, $\delta > 0$, and an infinite set $I \subseteq \mathbb{N}$ with $\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma_0))]_-\|_2 \geq \delta$ for $n \in I$. Fix a free ultrafilter ω on $\mathbb{N}$ with $I \in \omega$.

The adjoint model. Regard $M_{d_n}(\mathbb{C})$ as the Hilbert space $K_n = L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and let each group element act on it by conjugation, $\operatorname{Ad} U_n(g)\,\xi = U_n(g)\,\xi\,U_n(g)^*$. The operator-norm almost multiplicativity of U_n implies that $g \mapsto \operatorname{Ad} U_n(g)$ is an operator-norm asymptotic representation on these Hilbert spaces.

The ultraproduct. Let K_ω and $B_\omega = \prod_\omega B(K_n)$ be as in Section 3; write $[\,\cdot\,]_\omega$ for classes in either. The action of B_ω on K_ω is faithful, and B_ω is finite: every isometry in it is unitary. For projections in B_ω, inclusion of ranges in K_ω is equivalent to the projection order. Asymptotic multiplicativity implies that

$$\pi\colon H \longrightarrow \mathcal{U}(B_\omega), \qquad \pi(g) = [\operatorname{Ad} U_n(g)]_\omega,$$

a homomorphism.

The projection onto the fixed subspace. Choose a finite symmetric Kazhdan set $S \subseteq \Gamma$ containing 1, with Kazhdan constant $\kappa \in (0, 1]$. Let h be the Hermitian part of $\frac{1}{|S|}\sum_{s' \in S} \pi(\iota(s'))$ and let $\mathrm{Fix} \subseteq K_\omega$ be the subspace of vectors fixed by every $\pi(\iota(\gamma))$. Both Fix and its orthocomplement are invariant under each unitary $\pi(\iota(s'))$, hence under h. On Fix, $h = 1$. The restriction of $\pi \circ \iota$ to $\mathrm{Fix}^\perp$ has no nonzero invariant vector, so a unit vector $\xi \in \mathrm{Fix}^\perp$ satisfies $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$ for some generator. For that generator $\mathrm{Re}\langle \pi(\iota(s'))\xi, \xi\rangle \leq 1 - \kappa^2/2$, while every other term is at most 1. Averaging gives $\langle h\xi, \xi\rangle \leq 1 - \kappa^2/(2|S|)$. Consequently

$$\mathrm{sp}(h) \subseteq \Big[-1,\, 1 - \tfrac{\kappa^2}{2|S|}\Big] \cup \{1\},$$

and the spectral projection P of h at the isolated point 1 belongs to B_ω by continuous functional calculus and projects onto Fix.

The two projections agree. Let $V = \pi(s)$ and $Q = VPV^*$. Since $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$, every vector of Fix is fixed by each $\pi(s\iota(\gamma)s^{-1})$; and η is fixed by all $\pi(s\iota(\gamma)s^{-1})$ exactly when $V^*\eta \in \mathrm{Fix}$. Hence $\operatorname{ran} P = \mathrm{Fix} \subseteq V\,\mathrm{Fix} = \operatorname{ran} Q$, so $P \leq Q$. The element $r = V^*Q$ satisfies $r^*r = Q$ and $rr^* = V^*QV = P$, so $P \sim Q$. Since B_ω is finite, Lemma 3.1 gives $Q = P$.

Conclusion. Let $\xi = [x_n]_\omega \in K_\omega$; the uniform operator-norm bound makes this class well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each $\pi(\iota(\gamma))$ fixes ξ. Thus $\xi \in \mathrm{Fix}$. Since $Q = P$, we obtain $V\xi \in V\,\mathrm{Fix} = \operatorname{ran} Q = \operatorname{ran} P = \mathrm{Fix}$. The matrix $U_n(s)x_nU_n(s)^*$ is the vector $\operatorname{Ad} U_n(s)\,x_n$ of K_n, so, again by unitary invariance,

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]_-\|_2 = \big\|(\operatorname{Ad} U_n(\iota(\gamma)) - 1)\ \operatorname{Ad} U_n(s)\,x_n\big\| \longrightarrow 0$$

along ω, for every $\gamma \in \Gamma$. For $\gamma = \gamma_0$ this contradicts $I \in \omega$. $\square$

4. The central involution obstruction

Let w be a central involution of H and let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(H)$, and $q = \frac{1}{2}(1-\Theta(w))$ is a nonzero projection. Compressing the representation to q gives a matrix model of H in which the image of w is -1 and each block has its own normalized trace.

On the corner $q_n M_{d_n}(\mathbb{C}) q_n$ the operator norm is the restriction of the ambient one, so the compressed and polar-corrected maps remain operator-norm asymptotically multiplicative, while the trace is the corner's own normalized trace tr_{r_n}, not the restriction of tr_{d_n}. Hence $\|1-(-1)\|_{2,\mathrm{tr}_{r_n}} = 2$ for every value of r_n/d_n. Renormalizing the compressed corner is what protects the normalized Hilbert–Schmidt estimate from identity-block amplification: padding by an identity block leaves operator-norm defects and separations unchanged but can make normalized Hilbert–Schmidt distances arbitrarily small (Section S13).

Theorem 4.1 (central involution obstruction)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \qquad [c, \iota(\Gamma)] = 1.$$

Put $d = tct^{-1}$ and $u = [d, \iota(a)]$ for some $a \in \Gamma$. If $w := u^2$ is a nontrivial central involution of H, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps w to the identity, and H is not MF.

Proof. Let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism and suppose $\Theta(w) \neq 1$.

The (-1)*-spectral corner.* Choose coordinate unitary lifts $(V_{g,n})$ of Θ (Lemma 2.2) and put $p_n = \frac{1}{2}(1 + V_{w,n})$, the averaging operator of the two-element subgroup $F = \{1, w\}$. Because the lifts are asymptotically multiplicative and $w^2 = 1$, the Hermitian part of p_n is asymptotically idempotent. Its spectrum therefore concentrates near $\{0, 1\}$, and continuous functional calculus at $\frac{1}{2}$ gives projections at operator-norm distance $o(1)$ from p_n. Since w is central, these projections asymptotically commute with every lift of $\Theta(H)$. Let q_n be their complements. If q_n vanished for all large n the averages would converge to 1 and $\Theta(w)$ would be 1; so $q_n \neq 0$ along infinitely many coordinates, which we retain and relabel by $\mathbb{N}$. Compress each lift to $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$. Asymptotic centrality makes the compressed matrices almost unitary in the corner, so polar correction gives an operator-norm asymptotic representation $(W_{g,n})$ of H on nonzero blocks with

$$\|W_{w,n} + 1\| \longrightarrow 0.$$

Equip each corner block with its own normalized trace tr_{r_n}; all estimates below are relative to r_n and independent of the ratio r_n/d_n.

The commutant on the corner. Since c centralizes $\iota(\Gamma)$, the sequence $(W_{c,n})$ centralizes $(W_{\iota(\gamma),n})$ asymptotically in operator norm, hence in normalized

Hilbert–Schmidt norm. By Theorem 3.4 applied to $x_n = W_{c,n}$, the conjugated sequence $(W_{d,n})$ lies in the same asymptotic commutant. Therefore $\|W_{u,n} - 1\|_2 \to 0$, and squaring gives $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$.

The contradiction. The relation $w = u^2$ gives $\left\|W_{u,n}^2 - W_{w,n}\right\| \to 0$, while $W_{w,n} \to -1$ in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence $\left\|W_{u,n}^2 + 1\right\|_2 \to 0$ and $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$. The triangle inequality would then give $\|2 \cdot 1\|_2 \to 0$, contradicting $\|2 \cdot 1\|_2 = 2$. Thus $\Theta(w) = 1$. Since $w \neq 1$, every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ has nontrivial kernel, so H is not MF. $\square$

The finite-normal criterion extends the same argument to arbitrary finite normal subgroups of N_{conj} (Theorem S5.3). A second variant treats normal property-(T) subgroups $K \leq N_{\mathrm{conj}}$, with a Kazhdan spectral projection in place of the two-element average; there K may be infinite, noncentral and torsion-free (Theorem S5.6).

5. The Kazhdan–Clifford construction

Theorem 5.1 (Kazhdan–Clifford construction). *Let Γ be a finitely presented group with property* (T)*, let $\alpha\colon \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Adjoin elements t, c and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \qquad c^2 = 1, \qquad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$d = tct^{-1}, \qquad u = [d, a], \qquad w = u^2 = [d, ada^{-1}],$$

and impose centrality of w. More explicitly, for a finite generating set S_Γ it is enough to impose the conjugation and commutation relations for $s \in S_\Gamma$, together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \qquad [w, t] = [w, c] = 1.$$

The group $E(\Gamma, \alpha, a)$ is finitely presented, the canonical map $\Gamma \to E(\Gamma, \alpha, a)$ is injective, and w is a nontrivial central involution. Every homomorphism $E(\Gamma, \alpha, a) \to \mathcal{U}(\mathcal{Q})$ maps w to 1, so $E(\Gamma, \alpha, a)$ is not MF.

Proof. Finite presentability follows from the relative presentation. Start with the free product of Γ and the free group on t, c, and quotient by the finitely many displayed relations for a finite generating set of Γ.

The relation $w^2 = 1$ is redundant. Put $b = ada^{-1}$. Both d and b are involutions, and $w = [d, b] = (db)^2$. A direct computation gives

$$dwd^{-1} = w^{-1}. \tag{4}$$

Centrality also gives $dwd^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

concretely. Let Γ_∞ be the direct limit of the system $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots$, whose nth copy of Γ we call its *nth level*; injectivity of α passes to the limit, so the level-zero map embeds Γ in Γ_∞. Applying α levelwise induces an injective endomorphism σ of Γ_∞. It is surjective, hence an automorphism, because an element at level k equals σ of the same element at level $k+1$. Put $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$, with stable letter t implementing σ; then $t\gamma t^{-1} = \alpha(\gamma)$ for $\gamma \in \Gamma$. Write $X = G/\Gamma$ for the coset space of the level-zero copy. Let G permute the Clifford generators $(e_x)_{x\in X}$ and map c to e_Γ. Then

$$d \longmapsto e_{t\Gamma}, \qquad ada^{-1} \longmapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \longmapsto (e_{t\Gamma} e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows both that $w \neq 1$ and that the canonical map of Γ is injective.

Thus $w = u^2$ is a nontrivial central involution, and by Theorem 4.1 every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps w to the identity. Since $w \neq 1$, every such homomorphism has nontrivial kernel. □

6. A finitely presented non-MF group

Specialize Theorem 5.1 to

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I). \tag{5}$$

The affine group Γ is finitely presented and has property (T) [BHV, Example 1.7.4(i)], and Section S1 identifies it with the presentation used below. The endomorphism α is injective because doubling is injective on $\mathbb{Z}^3$, and it is not surjective because $e_1 \notin 2\mathbb{Z}^3$. The same observation gives $a \notin \alpha(\Gamma)$. Hence Theorem 5.1 applies. Relative to Γ, the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \;\middle|\; \begin{array}{c} t\gamma t^{-1} = \alpha(\gamma), \quad [c,\gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, \qquad [w, g] = 1 \end{array} \right\rangle,$$

where S_Γ is a finite generating set of Γ, g runs over the generators, and $w = [\,tct^{-1},\, a\,(tct^{-1})\,a^{-1}]$. Substituting a standard presentation of the affine group in the generators v_1, v_2, v_3 for the translations and x, y, z for the linear part yields the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$\mathcal{R} = \left\langle x, y, z \;\middle|\; \begin{array}{c} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle, \tag{6}$$

and the *base*

(7)

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \;\middle|\; \begin{array}{c} \text{the eight relations in (6)}, \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3,\ xv_2x^{-1} = v_1,\ xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1,\ yv_2y^{-1} = v_2^{-1}v_3,\ yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1},\ zv_2z^{-1} = v_1v_3^{-1},\ zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Write $\iota \colon \mathcal{B} \to E$ for the homomorphism induced by the first six generators of the presentation below.

Definition 6.1. Use the six base generators and the relations displayed in (7). Define

(8)

$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \;\middle|\; \begin{array}{c} \text{(B)} \quad \text{all relations in (7)}, \\ \text{(T)} \quad tv_it^{-1} = v_i^2\ (1 \le i \le 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C)} \quad c^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W)} \quad [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle,$$

where w abbreviates the word $\left[tct^{-1},\, v_1(tct^{-1})v_1^{-1}\right]$. This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). Family (B) presents the base. Family (T) states that conjugation by t maps each base generator into the base. Family (C) states that c is an involution commuting with the base generators, and family (W) states that w is central. Centrality together with $c^2 = 1$ implies $w^2 = 1$ by (4). Families (T) and (C) give, respectively,

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \qquad [c, \iota(\mathcal{B})] = 1.$$

The two relations imply that $d = tct^{-1}$ commutes with $t\iota(\mathcal{B})t^{-1}$. They impose no relation requiring d to commute with $\iota(v_1)$, which lies outside that conjugate subgroup; Figure 1 depicts this configuration.

A Tietze reduction removes two generators and nine relators (Section S1).

Lemma 6.2 (the distinguished word is a square)**.** *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$\left[\, d,\ a_1\, d\, a_1^{-1} \right] \;=\; \left[\, d, a_1 \,\right]^2.$$

Proof. Put $b = a_1da_1^{-1}$, so $b^2 = a_1d^2a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = da_1d^{-1}a_1^{-1} = d\,(a_1da_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db\,db = d\,b\,d^{-1}b^{-1} = [d, b]. \qquad \square$$

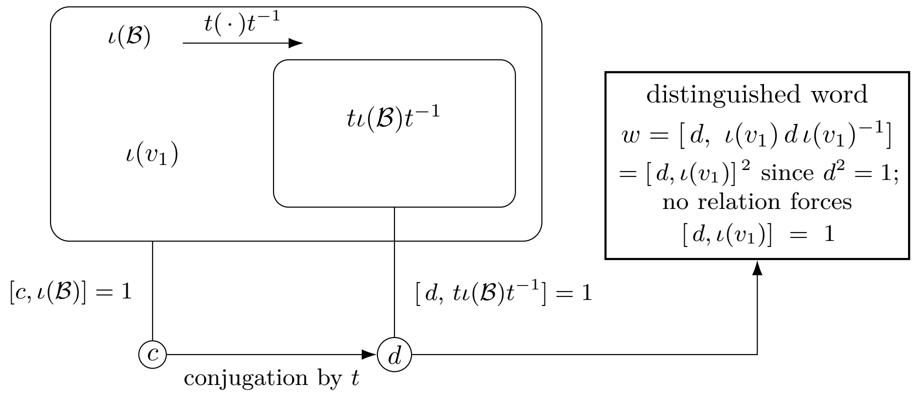

Figure 1: The one-sided conjugation configuration. The relations of (8) give $t\iota(\mathcal{B})t^{-1} \subsetneq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, hence $[\,d,\ t\iota(\mathcal{B})t^{-1}] = 1$ for $d = tct^{-1}$; the commutator $[\,d, \iota(v_1)]$ remains unconstrained, and $w = [\,d,\ \iota(v_1)d\iota(v_1)^{-1}] = [\,d, \iota(v_1)]^2$. Section S2 proves that the inclusion is strict and that $\iota(v_1)$ lies outside the conjugate subgroup.

Proof of Theorem A. Proposition S2.3 gives $w \neq 1$.

Let (d_n) be given, write $\mathcal{Q} = \mathcal{Q}((d_n))$, and let $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$ be a homomorphism. Apply the central involution obstruction (Theorem 4.1) with $\Gamma = \mathcal{B}$, the canonical homomorphism ι, the elements $t, c \in E$, and $a = v_1$. Proposition S1.1 gives property (T) of $\mathcal{B}$, while families (T) and (C) give $t\iota(\mathcal{B})t^{-1} \subsetneq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, respectively. In E we have $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 6.2 with $a_1 = \iota(v_1)$ gives

$$w = [\,d,\ \iota(v_1)\,d\,\iota(v_1)^{-1}] = [\,d, \iota(v_1)\,]^{\,2} = u^2, \qquad u = [\,d, \iota(v_1)\,].$$

The defining relations make w central, and $c^2 = 1$ then implies $w^2 = 1$ by (4). Proposition S2.3 gives $w \neq 1$. Hence $\Theta(w) = 1$.

Every homomorphism $E \to \mathcal{U}(\mathcal{Q})$ maps the nontrivial element w to 1, so none is injective and E is not MF. The canonical maps $E \to \mathcal{U}(C^*_{\mathrm{r}}(E))$ and $E \to \mathcal{U}(C^*_{\max}(E))$ are injective, the first because $g \mapsto \lambda_g$ is and the second because the left regular representation factors through $C^*_{\max}(E)$. If $e\colon A \to \mathcal{Q}$ is an injective, possibly nonunital, $*$-homomorphism from a unital algebra, then $u \mapsto e(u) + 1 - e(1)$ is an injective homomorphism $\mathcal{U}(A) \to \mathcal{U}(\mathcal{Q})$. Hence an MF embedding of either $C^*_{\mathrm{r}}(E)$ or $C^*_{\max}(E)$ would give an injective homomorphism $E \to \mathcal{U}(\mathcal{Q})$, a contradiction. $\square$

Corollary 6.3 (uniform finite operator-norm obstruction)**.** *There exist $\delta > 0$ and a finite set $F_0 \subset E$ such that, for every $d \geq 1$, every map $\phi\colon E \to \mathcal{U}(d)$ satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \qquad (g, h \in F_0)$$

satisfies $\|\phi(w) - 1\| < 1$.

Proof. Fix an exhaustion $F_1 \subseteq F_2 \subseteq \cdots$ of E by finite sets. If no such pair (δ, F_0) exists, then for each n there is a unitary-valued ϕ_n with multiplicative defect at most $1/n$ on F_n and $\|\phi_n(w) - 1\| \geq 1$. The sequence (ϕ_n) is an operator-norm asymptotic representation of E, so by Lemma 2.2 its class is a homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$, and $\|\Theta(w) - 1\| \geq 1$ gives $\Theta(w) \neq 1$, contradicting Theorem A. □

This argument is nonquantitative and gives no bound for δ or $|F_0|$. An effective modulus would need an effective Kazhdan constant for the base together with effective versions of the compression estimates.

7. Consequences of the example

Throughout, $d = tct^{-1}$ and $u = [d, \iota(v_1)]$, so that $w = u^2$. For a countable group G, write $\mathrm{Res}_{\mathrm{MF}}(G)$ for its *MF residual*: the intersection of the kernels of all its homomorphisms to $\mathcal{U}(\mathcal{Q})$ (Definition S9.1). Proposition S9.3 shows that G is MF exactly when this subgroup is trivial. By Theorem A, $1 \neq w \in \mathrm{Res}_{\mathrm{MF}}(E)$.

The quotient by the central involution.

Theorem B (the quotient of E by $\langle w \rangle$)**.** *The quotient $E/\langle w \rangle$ is not MF. The image of $[\iota(v_1), d] = u^{-1}$ is nontrivial in $E/\langle w \rangle$ and lies in $\mathrm{Res}_{\mathrm{MF}}(E/\langle w \rangle)$; consequently $\{1, w\}$ is a proper subgroup of $\mathrm{Res}_{\mathrm{MF}}(E)$.*

If an involution commutes with the conjugate copy of a Kazhdan subgroup and its conjugates under the original subgroup commute pairwise, then every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps all of those conjugates to the same element (Theorem S10.3). In E the pairwise commutators of those conjugates of d lie in $\langle w \rangle$, so in $E/\langle w \rangle$ the element d satisfies these hypotheses, and every homomorphism $E/\langle w \rangle \to \mathcal{U}(\mathcal{Q})$ maps u to 1. Thus $E/\langle w \rangle$ is a second finitely presented non-MF group, and $\mathrm{Res}_{\mathrm{MF}}(E)$ contains elements outside $\{1, w\}$ (Section S13).

A stably finite non-MF group algebra. Every MF algebra is stably finite [BK]; the converse fails. The canonical trace on $C^*_{\mathrm{r}}(E)$ is faithful, so $C^*_{\mathrm{r}}(E)$ is stably finite (Lemma S11.1). Abstract examples follow from the failure of the Connes embedding problem [FGH]; the reduced algebra of the finitely presented group E is an explicit one (Theorem C). For $C^*_{\mathrm{r}}(\mathrm{SL}_3(\mathbb{Z}))$ and $C^*_{\mathrm{r}}(\mathrm{SL}_4(\mathbb{Z}))$ the MF property is undecided; only the stronger PMF property is known to fail, for $\mathrm{SL}_4(\mathbb{Z})$ [MdlS].

Theorem C (the reduced group algebra of E)**.** *The algebra $C^*_{\mathrm{r}}(E)$ is unital and separable, has a faithful tracial state, and is therefore stably finite, but it is not an MF algebra.*

Proof. $C^*_{\mathrm{r}}(E)$ is unital and separable because E is countable, and by Lemma S11.1 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A. □

The algebra is also exact, by the block normal form of E and permanence for free products and extensions (Section S11). The algebra is not nuclear: the canonical map $\mathcal{B} \to E$ is injective (Section S2), so E contains an infinite Kazhdan group and is nonamenable, whence $C^*_{\mathrm{r}}(E)$ is not nuclear [Lan]. The nuclear form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear C^*-algebra is quasidiagonal — therefore remains open.

A sofic example.

Theorem D (a finitely presented sofic non-MF group)**.** *The group E of Definition 6.1 is finitely presented, sofic, hyperlinear, and non-MF. It is not LEF and therefore is not residually finite.*

Let E_0 be the kernel of the exponent sum in t. For every finite subset of E_0, the T-coordinates lie in a single level of the ascending chain and the lamp coordinates involve finitely many conjugates of c; the subgroup they generate embeds in a residually finite group. Hence E_0 is LEF (Section S11.3). Since E is a split extension of E_0 by $\mathbb{Z}$, it is sofic by the Elek–Szabó amenable-extension theorem [ES06]. This extension does not preserve LEF: LEF groups are MF [CDE], so E_0 is LEF whereas E is not.

Adding the anticommutation relations for every pair of distinct cosets gives a quotient map from E onto the Clifford group $W = \mathrm{Cl}(X) \rtimes G$ built in the proof of Theorem 5.1. This map sends w to the Clifford sign, which lies in $\mathrm{Res}_{\mathrm{MF}}(W)$ by Lemma S9.2, so W is not MF either. Its defining extension has locally finite kernel and residually finite quotient, both MF, so the MF property is not closed under extensions; and F_8 surjects onto E, so it is not closed under quotients (Section S11, Corollary S9.8).

Supplementary material

S1. The base, and a smaller presentation

The base $\mathcal{B}$ of (7) is the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, for which property (T) is classical [BHV, Example 1.7.4(i)].
Six generators suffice. The group E is generated by the six elements v_1, x, y, z, t, c, and has a Tietze-equivalent presentation on them with thirty-two relators. In (8), eliminate $v_3 = xv_1x^{-1}$ and $v_2 = x^2v_1x^{-2}$ by Tietze substitutions and delete the two defining relations; the remaining x-action relation follows from $x^3 = 1$, and the stable-letter, c-commutation, and centrality relations for v_2, v_3 follow from those for v_1 together with $[t, x] = 1$ and $x^3 = 1$. Deleting these nine leaves thirty-two relators on v_1, x, y, z, t, c.

After these Tietze transformations, the remaining relations still imply $[\,d,\ t\iota(\mathcal{B})t^{-1}] = 1$ for $d = tct^{-1}$ and impose no relation on $[\,d,\ \iota(v_1)\,d\,\iota(v_1)^{-1}]$.

S1.1. **Property (T) of the base.**

Proposition S1.1. *The twenty-relator group $\mathcal{B}$ has property* (T)*, in both the real-orthogonal and complex-unitary formulations.*

Proof. By Remark S1.2 the presentation (7) defines $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, which has property (T) [BHV, Example 1.7.4(i)]. For an orthogonal representation ρ on a real Hilbert space $H_{\mathbb{R}}$, the complexification $H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ carries the unitary representation $\rho \otimes 1$, whose invariant vectors and displacement norms are those of ρ in each of the two real coordinates; a Kazhdan pair for the unitary formulation is therefore one for the orthogonal formulation. □

Remark S1.2 (the classical identification). The eight relations (6) present $\mathrm{SL}_3(\mathbb{Z})$ [CRW, Theorem 2]; the corresponding matrices are also recorded in [CLV]. Hence $\mathcal{B} \cong \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, and the surjection $\mathcal{B} \to \bar{\Gamma}$ of Lemma S2.2 onto the matrix realization is an isomorphism.

S2. Nontriviality of w

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2].

Construction S2.1 (Clifford groups)**.** For a set X, use the real Clifford-algebra convention generated by $(c_x)_{x \in X}$ with $c_x^2 = 1$ and $c_xc_y = -c_yc_x$ for $x \neq y$, and let $\mathrm{Cl}(X)$ be the subgroup of units generated by -1 and the c_x. Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \qquad c_xc_y = \zeta c_yc_x \quad (x \neq y).$$

The sign ζ is nontrivial. Thus $\mathrm{Cl}(X)$ is a central extension of the elementary abelian 2-group on X, with central kernel generated by ζ. For distinct $x, y \in X$, the commutator $[c_x, c_y]$ equals ζ. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ.

Lemma S2.2 (linear model)**.** *For $M \in \mathrm{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M,u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$x \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), \qquad y \mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right),$$

$$z \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right).$$

Let $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2,2,2,1)$. The six matrices satisfy all twenty relators of $\mathcal{B}$, and

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z. Moreover the matrix assigned to v_1 does not lie in $\mathrm{range}(\bar{\alpha})$*.*

Proof. Each of the twenty relators reduces to an equality between explicit 4×4 rational matrices. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\mathrm{GL}_4(\mathbb{Z})$. Hence $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$. However $D^{-1}\bar{v}_1 D$ has entry $1/2$ in position $(1,4)$, so it is not in $\bar{\Gamma}$. This proves the last assertion. ◻

Form the direct limit $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$ and let τ denote its shift automorphism. Set $\Sigma = T \rtimes \mathbb{Z}$. This is the group G used in Theorem 5.1 for the data (5). Let $j \colon \bar{\Gamma} \hookrightarrow \Sigma$ be the level-zero map. Put $X = \Sigma / j(\bar{\Gamma})$, with base point $o = j(\bar{\Gamma})$. Since $D^{-1}\bar{v}_1 D \notin \bar{\Gamma}$ (Lemma S2.2), we have $\tau o \neq j(\bar{v}_1)\tau o$, and the Clifford generators at these two points anticommute.

Proposition S2.3. *$w \neq 1$ in E.*

Proof. Form $W := \mathrm{Cl}(X) \rtimes \Sigma$ as in Construction S2.1. Define a map on the generators of (8) by

$$g \mapsto j(\bar{g}) \in \Sigma \quad (g \in \{v_1, v_2, v_3, x, y, z\}),$$
$$t \mapsto \tau, \qquad c \mapsto c_o.$$

The exact checks in Lemma S2.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes o, so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \qquad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two points are distinct by Lemma S2.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W, the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \to W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E. □

The two points are distinct because $D^{-1}\bar{v}_1 D$ has the entry $1/2$ in position $(1, 4)$ and therefore does not lie in $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$.

S3. The MF equivalences

An *approximate unitary representation* for a countable group G, a finite set $F \subseteq G$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$ is a map $V \colon G \to \mathcal{U}(r)$, for some positive dimension r, such that

$$\|V(g)V(h) - V(gh)\| \leq \varepsilon \quad (g, h \in F),$$
$$\|V(g) - V(h)\| \geq \delta \quad (g \neq h \in F).$$

The local formulation in Proposition 2.3 asks for such a model with $\delta = 1$ for every finite F and every $\varepsilon > 0$.

Proof of Proposition 2.3. Lemma 2.2 identifies $\prod_n \mathcal{U}(d_n)/\mathcal{N}$ with $\mathcal{U}(\mathcal{Q})$, which is the first equivalence once the dimension convention is normalized.

Dimension normalization. Given positive dimensions d_n, replace the nth sequence of unitary lifts by

$$\widetilde{V}_n(g) = I_{\mathbb{C}^{d_0}} \oplus \cdots \oplus I_{\mathbb{C}^{d_{n-1}}} \oplus V_n(g)$$

on $\mathbb{C}^{d_0} \oplus \cdots \oplus \mathbb{C}^{d_n}$, of strictly increasing dimension $D_n = d_0 + \cdots + d_n$. Block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements, so for all g, h

$$\left\|\widetilde{V}_n(g)\widetilde{V}_n(h) - \widetilde{V}_n(gh)\right\| = \|V_n(g)V_n(h) - V_n(gh)\|,$$
$$\left\|\widetilde{V}_n(g) - \widetilde{V}_n(h)\right\| = \|V_n(g) - V_n(h)\|.$$

The defect and separation data are unchanged. Hence the padded lifts define a homomorphism to $\mathcal{U}(\mathcal{Q}((D_n)))$, and it separates exactly the same group elements as the original homomorphism. The converse is immediate.

Suppose first that $\rho \colon G \to \mathcal{U}(\mathcal{Q}((d_n)))$ is injective, and choose unitary lifts $u_n(g)$ of $\rho(g)$. Fix a finite set $F \subseteq G$ and $\varepsilon > 0$. For each ordered pair $g \neq h$ in F, injectivity says that $(u_n(g)^{-1}u_n(h))_n$ does not converge to the identity in operator norm. Hence there are $\delta_{g,h} > 0$ and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer $N_{g,h}$ with $8 < N_{g,h}\,\delta_{g,h}^2$. The following estimate then applies: if $u, v \in \mathcal{U}(d)$ satisfy $\|u - v\| \geq \delta$, then $\|u^{\otimes p} - v^{\otimes p}\| \geq 1$ for some $p \leq N$ whenever $8 < N\delta^2$. Indeed, put $z = u^*v$, so that $\|u - v\| = \|1 - z\|$

and z has a spectral value $e^{2\pi i\alpha}$ with $|1-e^{2\pi i\alpha}| = 2\sin(\pi\alpha') \geq \delta$, where $\alpha' = \min(\{\alpha\}, 1-\{\alpha\})$; since $(u^{\otimes p})^* v^{\otimes p} = z^{\otimes p}$ has $e^{2\pi i p\alpha}$ in its spectrum,

$$\left\|u^{\otimes p} - v^{\otimes p}\right\| \ \geq\ \left|1-e^{2\pi i p\alpha}\right| \ =\ 2\sin(\pi \min(\{p\alpha\}, 1-\{p\alpha\})),$$

which is at least 1 whenever $\{p\alpha\}$ lies in the closed interval $[1/6, 5/6]$; the endpoints are included because they give separation exactly 1. The bound $2\sin(\pi\alpha') \geq \delta$ gives $\alpha' \geq \delta/(2\pi)$; consecutive multiples of α move $\{p\alpha\}$ by α' in a fixed direction modulo 1, which is less than the length $2/3$ of that interval, so some $p \leq \lceil 1/(6\alpha')\rceil \leq \pi/(3\delta) + 1 < N$ lands in it. Because ρ is a homomorphism into the quotient, all multiplicative defects for pairs in F are arbitrarily small at every sufficiently large coordinate. Choose, for each $g \neq h$, a coordinate with the displayed separation and with every required defect at most $\varepsilon/N_{g,h}$. Then the pth tensor power separates the designated pair by at least 1, while the tensor-product telescoping identity bounds each new defect by $p\,\varepsilon/N_{g,h} \leq \varepsilon$. Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in F is separated by at least 1 and every required product has defect at most ε. This gives the normalized approximate representation. If $|F| \leq 1$ there are no distinct pairs and hence no blocks, and the dimension is required to be positive; in that case take $r = 1$ and the map sending every element of G to $1 \in \mathcal{U}(1)$, which is exactly multiplicative and separates the empty set of pairs.

Conversely, suppose approximate unitary representations with separation constant 1 exist for all finite sets and tolerances. Choose models along an exhaustion with errors tending to zero. For each fixed pair g, h, the multiplicative defect tends to 0, and for each fixed pair $g \neq h$, the separation is eventually at least 1. Hence the classes define an injective homomorphism into $\prod_n \mathcal{U}(r_n)/\mathcal{N}$, with no condition on the dimension sequence. In particular every nonidentity element is separated by some homomorphism to a matrix quotient, so the MF residual of Definition S9.1 is trivial, and Proposition S9.3(2) then combines these separating sequences into a single faithful homomorphism to $\mathcal{U}(\mathcal{Q})$.

□

S4. Weights and intertwiners

Theorem 3.4 holds with an arbitrary weight ν_n in place of the dimension, and between two asymptotic representations in place of one; the theorem as stated is the case $\nu_n = d_n$.

Theorem S4.1 (one-sided conjugation at an arbitrary weight, and for intertwiners)**.** *Let* Γ*,* H*,* ι*, and* s *be as in Theorem 3.4.*

(1) Let (U_n) *be as there, let* (ν_n) *be nonnegative weights, and suppose there is* C *with* $\mathrm{Tr}(x_n^* x_n) \le C\,\nu_n$ *for all* n*, and that for every* $\gamma \in \Gamma$ *and every* $\varepsilon > 0$*, eventually*

$$\mathrm{Tr}\,\left|x_n - U_n(\iota(\gamma))\,x_n\,U_n(\iota(\gamma))^*\right|^2 \le \varepsilon\,\nu_n.$$

Then $U_n(s)\,x_n U_n(s)^*$ *and* $U_n(s)^*\,x_n U_n(s)$ *satisfy both conditions as well.*

(2) Let $U_n^1\colon H \to \mathcal{U}(d_n^1)$ *and* $U_n^2\colon H \to \mathcal{U}(d_n^2)$ *be operator-norm asymptotic representations and let* (x_n) *be uniformly operator-norm bounded* $d_n^1 \times d_n^2$ *matrices with*

$$\frac{\mathrm{Tr}\,\left|x_n - U_n^1(\iota(\gamma))\,x_n\,U_n^2(\iota(\gamma))^*\right|^2}{d_n^1 + d_n^2} \longrightarrow 0 \qquad (\gamma \in \Gamma).$$

Then $U_n^1(s)\,x_n\,U_n^2(s)^*$ *satisfies the same convergence for every* $\gamma \in \Gamma$*.*

Proof. (1) For $\nu_n > 0$, equip the nth coordinate space with the inner product $\langle x, y\rangle = \mathrm{Tr}(y^* x)/\nu_n$; an index with $\nu_n = 0$ forces $x_n = 0$. Conjugation by a unitary remains unitary for this inner product. The bound $\mathrm{Tr}(x_n^* x_n) \le C\nu_n$ makes (x_n) a bounded ultraproduct vector, and the adjoint actions remain operator-norm almost multiplicative because the weight does not enter the operator norm. Thus the ultraproduct proof of Theorem 3.4 applies with tr_{d_n} replaced by $\mathrm{Tr}(\,\cdot\,)/\nu_n$. Finiteness of the norm ultraproduct again turns $P \le VPV^*$ into $P = VPV^*$, which gives both conjugation directions.

(2) The block sum $U_n^\oplus = U_n^1 \oplus U_n^2$ is an operator-norm asymptotic representation, since block-diagonal operator norms are maxima. Embed x_n in the off-diagonal corner of the block sum; this embedding preserves the operator norm. Its commutator with $U_n^\oplus(\iota(\gamma))$ is the off-diagonal embedding of the intertwining defect, whose normalized Hilbert–Schmidt norm on the block space has denominator $d_n^1 + d_n^2$. Conjugation by $U_n^\oplus(s)$ acts on the embedded corner by $x_n \mapsto U_n^1(s)\,x_n\,U_n^2(s)^*$. Theorem 3.4 applied to $U_n^\oplus$ gives the conclusion for the embedded sequence. Replacing the denominator $d_n^1 + d_n^2$ by an arbitrary weight ν_n, and applying (1) to the block sum in place of Theorem 3.4, gives (2) at every weight. □

Taking $\nu_n = d_n$ gives Theorem 3.4 with the uniform operator-norm bound relaxed to a uniform bound on $\mathrm{Tr}(x_n^* x_n)/d_n$. Taking $\nu_n = k_n$, the rank of the projection lift in the proof of Theorem S10.1, gives the normalization used there.

Invariant tensors of type (p,q) may be identified with Hilbert–Schmidt intertwiners from the qth to the pth tensor power. Since $\|A^{\otimes p} - B^{\otimes p}\| \le p\,\|A - B\|$, tensor powers of an operator-norm asymptotic representation are again operator-norm asymptotic representations. Theorem S4.1(2) therefore applies to every pair of tensor powers: at each fixed tensor type and each weight, a sequence asymptotically invariant for the conjugate copy of the Kazhdan image is asymptotically invariant for the whole of it.

S5. Obstructions from a one-sided conjugation datum

Each result below combines 2-norm triviality of a subgroup of H with a corner on which that subgroup stays a positive 2-norm distance from the identity, so the subgroup lies in the kernel of every homomorphism $H \to \mathcal{U}(\mathcal{Q})$. The source of the corner differs:

- Theorem S5.3: a *finite* normal subgroup of N_{conj}, the corner being the complement of its rounded averaging operator — the case of the central involution of Section 4;
- Theorem S5.6: a normal property-(T) subgroup of N_{conj}, which may be infinite, noncentral and torsion-free, the corner being a spectral projection of a Kazhdan average;
- Theorem S5.7: a version that assumes 2-norm triviality directly, without a conjugation datum;
- Corollary S5.8: that statement applied to the subgroup $\mathfrak{D}(H, L)$ of (9), which depends on L alone and not on a choice of t and c.

Definition S5.1 (Kazhdan conjugation datum)**.** Let H be a countable group. A *Kazhdan conjugation datum* in H is a tuple (Λ, ι, t, c), where Λ is a countable group with property (T), $\iota\colon \Lambda \to H$ is a group homomorphism, possibly with nontrivial kernel, and $t, c \in H$, subject to

(M1) $t\,\iota(\lambda)\,t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;
(M2) $c\,\iota(\lambda) = \iota(\lambda)\,c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The datum determines the normal closure

$$N_{\mathrm{conj}} \;=\; \big\langle\!\big\langle\, [\,d, \iota(\lambda)\,] \;:\; \lambda \in \Lambda \,\big\rangle\!\big\rangle_H \;\trianglelefteq\; H.$$

For a finitely generated Λ, conditions (M1)–(M2) are imposed by finitely many relations once generating sets are fixed. Conditions (M1)–(M2) imply that c commutes with $\iota(\Lambda)$ and that d commutes with the subgroup $t\iota(\Lambda)t^{-1}$. The commutators $[d, \iota(\lambda)]$ remain unrestricted, and N_{conj} is their normal closure.

Definition S5.2 (2-norm triviality)**.** Let $U = (U_n)$ be an operator-norm asymptotic representation of a countable group H and let ω be an ultrafilter refining the cofinite filter. Write

$$K_2^\omega(U) \;=\; \big\{ g \in H :\; \lim_\omega \|U_n(g) - 1\|_2 = 0 \big\},$$

the kernel of the homomorphism induced by U into the tracial ultraproduct of the matrix blocks along ω, and hence a normal subgroup of H. An element g is 2-*norm trivial* if $g \in K_2^\omega(U)$ for every such U and every such ω. If $\|U_n(g)-1\|_2 \to 0$ along the full sequence, then $g \in K_2^\omega(U)$ for every such ω.

Theorem S5.3 (finite-normal obstruction criterion)**.** *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c), and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\mathrm{conj}}$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

Proof. 2-*norm triviality of* N_{conj}. Let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H — in particular, coordinate unitary lifts of a homomorphism to $\mathcal{U}(\mathcal{Q})$ — write $U_{g,n} = U_n(g)$, and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \longrightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. Since c centralizes $\iota(\Lambda)$, the sequence $(U_{c,n})$ belongs to the asymptotic commutant of $\iota(\Lambda)$ in normalized Hilbert–Schmidt norm. Theorem 3.4, applied to $x_n = U_{c,n}$, places $(U_{d,n})$ for $d = tct^{-1}$ in the same asymptotic commutant. Therefore $[d, \iota(\lambda)] \in K_2$ for every $\lambda \in \Lambda$; since K_2 is normal, it contains the normal closure of these commutators: $N_{\mathrm{conj}} \subseteq K_2$, and so $F \subseteq K_2$. Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of N_{conj} is 2-norm trivial (Definition S5.2).

The coordinate averaging corner. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on F; choose coordinate unitary lifts $(V_{g,n})$ of Θ and form the coordinate averaging operators

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and F is finite, (p_n) is asymptotically self-adjoint and idempotent. Since F is normal, (p_n) asymptotically commutes with every lift of $\Theta(H)$. Continuous functional calculus applied to the Hermitian parts yields genuine projections at operator-norm distance $o(1)$ from p_n, and these projections are asymptotically central as well. Let q_n be their complements. Nontriviality of Θ on F forces $q_n \neq 0$ along infinitely many coordinates — otherwise the averages converge to 1 and $\Theta|_F$ is trivial. Retain exactly those coordinates and relabel them by $\mathbb{N}$. Compress every lift to the corner $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ with $r_n = \operatorname{rank} q_n > 0$. Asymptotic centrality implies that each compression $q_n V_{g,n} q_n$ is almost unitary in the corner, so polar correction changes it by $o(1)$ in operator norm and preserves asymptotic multiplicativity. The result is an operator-norm asymptotic representation $(W_{g,n})$ of H on the nonzero corner blocks. Since the corner is the complement of the rounded average, the finite average vanishes there:

$$(*) \qquad \Big\| \sum_{f \in F} W_{f,n} \Big\| \longrightarrow 0 \quad \text{in operator norm.}$$

The contradiction. Since $F \subseteq N_{\mathrm{conj}}$ and $(W_{g,n})$ is itself an operator-norm asymptotic representation of H, every $f \in F$ satisfies $\|W_{f,n} - 1\|_2 \to 0$. Because F is finite, these convergences are uniform over $f \in F$, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block with its own normalized trace. This contradicts $(*)$, because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ is trivial on F. □

The same conclusion holds with N_{conj} replaced by any normal subgroup all of whose elements lie in $K_2^\omega(U)$ for every operator-norm asymptotic representation U of H and every ω; Theorem S5.5 is that statement for $\mathfrak{D}(H, L)$.

The central involution obstruction (Theorem 4.1) is the case $F = \{1, w\}$, where $p_n = \frac{1}{2}(1 + V_{w,n})$ and the complementary corner gives a model in which $w = -1$ exactly.

S5.1. **An intrinsic form.** For a subgroup $L \leq H$, let $G^+(L)$ be generated by the elements s satisfying $sLs^{-1} \subseteq L$ — the *one-sided conjugators* of L — and put

$$\mathfrak{D}(H, L) = \left\langle\!\!\left\langle [gzg^{-1}, \ell] : g \in G^+(L),\ z \in C_H(L),\ \ell \in L \right\rangle\!\!\right\rangle_H, \tag{9}$$

which depends only on the pair (H, L), not on a choice of datum.

Corollary S5.4 (one-sided conjugators preserve the commutant under conjugation and inverse conjugation)**.** *Let* (r_n) *and* $V_{g,n} \in \mathcal{U}(r_n)$ *be an operator-norm asymptotic representation of* H*, let* Λ *have property* (T)*, let* $\iota\colon \Lambda \to H$ *be a homomorphism, and put* $L = \iota(\Lambda)$*. Suppose that* $(x_n \in M_{r_n}(\mathbb{C}))$ *is uniformly bounded in operator norm and belongs to the asymptotic commutant in normalized Hilbert–Schmidt norm of* L*:*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \qquad (\lambda \in \Lambda).$$

Then for every $g \in G^+(L)$*, conjugation and inverse conjugation both preserve that asymptotic commutant. Explicitly, both*

$$V_{g,n} x_n V_{g,n}^* \quad \text{and} \quad V_{g,n}^* x_n V_{g,n}$$

satisfy the preceding convergence for every $\lambda \in \Lambda$*.*

Proof. For a one-sided conjugator s, Theorem 3.4 gives the conclusion for $V_{g,n} x_n V_{g,n}^*$, and the equality $P = VPV^*$ established in its proof gives $V^*PV = P$, hence $V^* \operatorname{Fix} = \operatorname{Fix}$ and the conclusion for $V_{g,n}^* x_n V_{g,n}$. The set of $g \in H$ for which conjugation by $V_{g,n}$ and by $V_{g,n}^*$ both preserve the asymptotic commutant is a subgroup, and it contains every one-sided conjugator; hence it contains $G^+(L)$. □

Theorem S5.5 (the subgroup $\mathfrak{D}$ and the MF residual)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $F \trianglelefteq H$ is finite and $F \le \mathfrak{D}(H, L)$, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps every element of F to the identity.*

Proof. Let $(U_{h,n})$ be coordinate lifts of an operator-norm asymptotic representation of H and put $K_2 = \{h : \|U_{h,n} - 1\|_2 \to 0\}$, a normal subgroup of H. If $z \in C_H(L)$, then $(U_{z,n})$ lies in the asymptotic commutant of L, and by Corollary S5.4 so does $(U_{gzg^{-1},n})$ for every $g \in G^+(L)$. Hence each generator $[gzg^{-1}, \ell]$, $\ell \in L$, displayed in (9) belongs to K_2; since $K_2 \trianglelefteq H$, it contains their normal closure: $\mathfrak{D}(H, L) \le K_2$. Every element of $\mathfrak{D}(H, L)$ is therefore 2-norm trivial in every operator-norm asymptotic representation of H. If a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ were nontrivial on a finite normal $F \le \mathfrak{D}(H, L)$, the averaging-corner argument of Theorem S5.3 would contradict this 2-norm triviality. □

S5.2. **The normal-Kazhdan form of the obstruction.** A finite group has property (T), and its averaging operator is the projection onto its fixed space. The spectral projection of a Kazhdan average replaces that operator for an arbitrary normal property-(T) subgroup, which may be infinite, noncentral and torsion-free.

Theorem S5.6 (normal-Kazhdan obstruction)**.** *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c), and let $K \trianglelefteq H$ be a normal subgroup with property* (T) *such that $K \subseteq N_{\mathrm{conj}}$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Since finite groups have property (T), Theorem S5.3 is the finite case of Theorem S5.6.

Proof. The containment $N_{\mathrm{conj}} \subseteq K_2$ from the proof of Theorem S5.3 holds for any normal subgroup, so every element of K is 2-norm trivial in every operator-norm asymptotic representation of H.

Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K. Its image $\bar{H} = \Theta(H)$ is countable, hence MF, and Proposition 2.3, applied along an exhaustion of $\bar{H}$, gives an operator-norm asymptotic representation $(V_{g,n})$ in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models provide operator-norm separation, but at growing dimension this yields no lower bound on normalized Hilbert–Schmidt distances. The datum and the subgroup pass to their Θ-images with normality and property (T) preserved, and $\bar{K} = \Theta(K)$ is nontrivial and lies in the subgroup N_{conj} determined by the image datum. We now work in $\bar{H}$.

The Kazhdan corner. Fix a free ultrafilter ω, let H_ω be the Hilbert-space ultraproduct of the coordinate spaces $\mathbb{C}^{d_n}$, and let $B_\omega = \prod_\omega B(\mathbb{C}^{d_n})$ act on it as in Section 3, so that the classes $\pi(g) = [V_{g,n}]_\omega$ define a unitary representation of $\bar{H}$ on H_ω. Fix a finite symmetric generating Kazhdan set

$S \subseteq \bar{K}$ with constant ε_0, put $\theta = 1 - \varepsilon_0^2/(4|S|)$, and let

$$h = \frac{1}{|S|}\sum_{a\in S}\pi(a) \in B_\omega,$$

self-adjoint because S is symmetric. A unit vector ξ has $\langle h\xi,\xi\rangle = 1$ exactly when $\pi(a)\xi = \xi$ for every $a \in S$, while the Kazhdan pair bounds $\langle h\xi,\xi\rangle$ by θ on the orthogonal complement of $\operatorname{Fix}\pi(\bar{K})$. The spectrum of h therefore lies in $[-1,\theta]\cup\{1\}$, the spectral projection $P = \chi_{\{1\}}(h)$ belongs to B_ω, and its range is $\operatorname{Fix}\pi(\bar{K})$. Put $q = 1 - P$.

If $q = 0$ then $\pi(k) = 1$ for every $k \in \bar{K}$, against the operator-norm separation of a fixed nontrivial element of $\bar{K}$. For $g \in \bar{H}$ the projection $\pi(g)P\pi(g)^*$ has range $\operatorname{Fix}\pi(g\bar{K}g^{-1}) = \operatorname{Fix}\pi(\bar{K})$ by normality, so P and q commute with $\pi(\bar{H})$ exactly. Lift q to projections $q_n \in M_{d_n}(\mathbb{C})$, nonzero along ω and with $\|[q_n, V_{g,n}]_-\| \to_\omega 0$ for every $g \in \bar{H}$.

To obtain ordinary convergence before compressing, pass to a subsequence. Enumerate $\bar{H} = \{g_1, g_2, \dots\}$. For each j the set

$$A_j = \{n \ : \ q_n \neq 0 \ \text{ and } \ \|[q_n, V_{g_i,n}]_-\| < 1/j \ \text{ for all } i \leq j\}$$

is a finite intersection of sets in ω, hence lies in ω and in particular is infinite. Choose $n_1 < n_2 < \cdots$ with $n_j \in A_j$ and relabel along $(n_j)_j$. On the relabelled sequence $q_n \neq 0$ for every n and $\|[q_n, V_{g,n}]_-\| \to 0$ in the ordinary sense for every fixed $g \in \bar{H}$, since $g = g_i$ lies in the constraint of A_j for all $j \geq i$. Compressing every model to $q_n M_{d_n}(\mathbb{C}) q_n$ and polar-correcting now yields an operator-norm asymptotic representation $(W_{g,n})$ of $\bar{H}$ on nonzero blocks.

A 2-norm separation on the corner. Equip each corner block with its renormalized trace. The corner compression satisfies $qhq \leq \theta q$, so at ω-most coordinates the average over S of the real parts of the renormalized traces of the compressed models is at most $\theta + o(1)$. For each n, at least one generator in S therefore has compressed model with real renormalized trace at most $\theta + o(1)$, and so normalized Hilbert–Schmidt distance at least $\sqrt{2(1-\theta)} - o(1)$ from the corner identity. Since S is finite, a subsequence realizes this at a single generator s_0.

The contradiction. The corner models form an operator-norm asymptotic representation of $\bar{H}$, and $\bar{K}$ lies in the subgroup N_{conj} determined by the image datum. Hence every element of $\bar{K}$, in particular s_0, is 2-norm trivial in the corner models: $\|W_{s_0,n} - 1\|_2 \to 0$, contradicting the positive lower bound above. □

Theorem S5.7 (obstruction from 2-norm triviality)**.** *Let H be a countable group and let $D \leq H$ be a subgroup every element of which is 2-norm trivial (Definition S5.2). Let $K \trianglelefteq H$ be a normal subgroup with property* (T) *such that $K \leq D$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Proof. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K. The image $\bar{H} = \Theta(H)$ is countable and MF, and Proposition 2.3 provides an operator-norm asymptotic representation of $\bar{H}$ separating every fixed nontrivial element. Every operator-norm asymptotic representation of $\bar{H}$ composes with Θ to one of H, so every element of $\Theta(D)$ is again 2-norm trivial, while $\bar{K} = \Theta(K) \le \Theta(D)$ is normal, Kazhdan and nontrivial. Repeat the Kazhdan-corner construction from the proof of Theorem S5.6 for $\bar{K}$: it produces a corner and an element $s_0 \in \bar{K}$ that stays a positive 2-norm distance from the corner identity. Since $s_0 \in \Theta(D)$ is 2-norm trivial, this is a contradiction. $\square$

Every element of $\mathfrak{D}(H, L)$ is 2-norm trivial, so Theorem S5.7 gives the following normal-Kazhdan analogue of Theorem S5.5:

Corollary S5.8 (normal-Kazhdan form of the intrinsic residual)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $K \trianglelefteq H$ is a normal subgroup with property* (T) *and $K \le \mathfrak{D}(H, L)$, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps every element of K to the identity.*

Proof. By the proof of Theorem S5.5, every element of $\mathfrak{D}(H, L)$ is 2-norm trivial in every operator-norm asymptotic representation of H. Theorem S5.7 with $D = \mathfrak{D}(H, L)$ concludes. $\square$

The subgroup generated by *all* normal property-(T) subgroups of $\mathfrak{D}(H, L)$ therefore also lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$.

S6. The finite-dimensional obstruction

Every finite-dimensional representation of E over any field maps w to the identity (Theorem 3.3 with $H = E$). Consequently finite-dimensional linear representations do not separate the points of E, and every finite quotient kills w. Since the Clifford quotient of Section S2 sends w to a nontrivial element, that quotient is infinite. The central involution w lies in the kernel of every homomorphism to a finite group, so E is a group of *Deligne type* in the terminology of [BDL, Definition 1.2]. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and E is such a counterexample for the operator norm.

For the subgroup-valued assertion of Theorem 3.3, let C be the commutant of $\pi(\Gamma)$. The elements $g \in H$ for which conjugation by $\pi(g)$ preserves C in both directions form a subgroup. The finite-dimensional argument above shows that every one-sided conjugator belongs to this subgroup, hence so does all of $G^+(\Gamma)$. Thus, for $g \in G^+(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence every displayed generator in (9) lies in $\ker \pi$; since $\ker \pi \trianglelefteq H$, the whole normal closure $\mathfrak{D}(H, \Gamma)$ does as well.

Corollary S6.1. *The group E of Definition 6.1 admits no injective homomorphism into* $\mathrm{GL}(V)$ *for any finite-dimensional vector space V over any field. More strongly, every finite-dimensional linear representation maps w to the identity, and every homomorphism to a finite group maps w to the identity.*

Proof. Apply Theorem 3.3 with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1; the hypotheses hold by the defining relations (8). Every finite-dimensional representation maps w to the identity, and $w \neq 1$ by Proposition S2.3. For a finite target, compose with its faithful left regular representation over $\mathbb{Q}$. The composite is a finite-dimensional linear representation and hence maps w to the identity; faithfulness of the left regular representation then implies that the original homomorphism also maps w to the identity. $\square$

S7. The cyclic comparison

Theorem S7.1 (the cyclic comparison). *Define the presentation*

$$E_{\mathrm{BS}} = \left\langle \gamma_0, t, c \;\middle|\; \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\mathrm{BS}}^2 = 1, \quad [w_{\mathrm{BS}}, \gamma_0] = [w_{\mathrm{BS}}, t] = [w_{\mathrm{BS}}, c] = 1 \end{array} \right\rangle,$$

where $w_{\mathrm{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. *Then* E_{BS} *is finitely presented and* $w_{\mathrm{BS}} \neq 1$. *There is an explicit surjective homomorphism from* E_{BS} *onto the subgroup of the Clifford semidirect product generated by the displayed images, and it maps* w_{BS} *to the nontrivial central sign. Nevertheless every finite-dimensional representation of* E_{BS} *over every field maps* w_{BS} *to the identity.*

Proof of Theorem S7.1. The relator $w_{\mathrm{BS}}^2 = 1$ is redundant: tct^{-1} and $\gamma_0(tct^{-1})\gamma_0^{-1}$ are involutions, so centrality forces it by (4). Deleting it gives a six-relator presentation of the same group. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. By the commutant argument of Theorem 3.3, every finite-dimensional representation over every field maps w_{BS} to the identity.

For nontriviality, use the affine realization of $\mathrm{BS}(1,2)$ and its coset action. The base point is fixed by the cyclic base, while the two points represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these points and map c to the generator at the base point. All seven relators hold, and

$$w_{\mathrm{BS}} \longmapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\mathrm{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjection onto a Clifford quotient. $\square$

Sharpness of the Kazhdan hypothesis. The Clifford attachment of the preceding proof already supplies the representation this needs, once it is run over finite site sets. Doing so gives a sequence of finite-dimensional monomial models of the seven relators in which w_{BS} acts by the scalar -1, and assembling them gives $\Theta\colon E_{\mathrm{BS}} \to \mathcal{U}(\mathcal{Q})$ with $\Theta(w_{\mathrm{BS}}) \neq 1$, its image MF because it sits inside $\mathcal{U}(\mathcal{Q})$ by construction. The Clifford quotient in Theorem S7.1 is moreover amenable — it is a subgroup of the semidirect product of the locally finite Clifford group on the coset space by the solvable affine group $\mathrm{BS}(1,2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ — but the construction above does not use that, and so needs no quasidiagonality. Thus the property (T) hypothesis in the central involution obstruction is essential. Equivalently, in the residual language of Section S9, $w_{\mathrm{BS}} \notin \mathrm{Res}_{\mathrm{MF}}(E_{\mathrm{BS}})$.

The preceding homomorphism to $\mathcal{U}(\mathcal{Q})$ does not arise from exact finite-dimensional representations. In a finite image, the doubling relation gives $\pi(\langle \gamma_0^2 \rangle) = \pi(\langle \gamma_0 \rangle)$; since $\pi(tct^{-1})$ centralizes the first subgroup, it centralizes $\pi(\gamma_0)$, and the image of w_{BS} is the identity. To transfer this obstruction to exact local embeddings, take a finite set containing the generators together with every intermediate prefix of every defining word and of w_{BS}. Any exact local embedding of this set evaluates those words correctly and therefore satisfies the same equations. Hence neither E_{BS} nor its Clifford quotient is LEF, and the latter is not residually finite.

S8. The scaling family

For $m \geq 2$, define E_m by replacing the three relations $tv_it^{-1} = v_i^2$ in the eight-generator presentation of E with $tv_it^{-1} = v_i^m$ and leaving every other relator unchanged. Then $E_2 = E$. Write w_m for the image of the same distinguished word w.

Corollary S8.1 (the scaling family)**.** *For every natural number $m \geq 2$, the group E_m is finitely presented and w_m is a nontrivial central involution mapped to the identity by every homomorphism $E_m \to \mathcal{U}(\mathcal{Q})$. Consequently E_m is not MF, and neither $C^*_{\max}(E_m)$ nor $C^*_{\mathrm{r}}(E_m)$ is an MF C^*-algebra.*

Proof. Replace the affine dilation $\operatorname{diag}(2, 2, 2, 1)$ by $\operatorname{diag}(m, m, m, 1)$ (both acting as in Lemma S2.2). It conjugates each integral translation v_i to v_i^m and fixes the three linear generators. Conjugating v_1 by the inverse dilation gives translation by e_1/m, which is not integral. Hence the two relevant cosets remain distinct, and the Clifford representation sends w_m to -1. The Kazhdan and central-involution arguments are unchanged for every $m \geq 2$. $\square$

S9. The MF residual

Definition S9.1. For a countable group G_1, the *MF residual* is

$$\mathrm{Res}_{\mathrm{MF}}(G_1) \;=\; \bigcap \{\ker\Theta :\ \Theta \text{ a homomorphism } G_1 \to \mathcal{U}(\mathcal{Q})\},$$

a normal subgroup of G_1. If G_1/N is MF then composing $G_1 \to G_1/N$ with an injective homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$ gives a homomorphism with kernel N, so $\mathrm{Res}_{\mathrm{MF}}(G_1) \leq N$. Thus the MF residual is contained in every normal subgroup whose quotient is MF, and Proposition S9.3 shows that its own quotient is MF. This is the sense in which *residual* is used here, as for the nilpotent and finite residuals. The intersection is the same whether the dimension sequences are required to be strictly increasing or allowed to be arbitrary positive (Proposition 2.3). This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

Lemma S9.2 (functoriality)**.** *For every homomorphism $\varphi\colon G_1 \to G_2$ of countable groups,*

$$\varphi(\mathrm{Res}_{\mathrm{MF}}(G_1)) \;\subseteq\; \mathrm{Res}_{\mathrm{MF}}(G_2).$$

In particular, if $\varphi\colon E \to G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

Proof. If $x \in \mathrm{Res}_{\mathrm{MF}}(G_1)$ and Θ is a homomorphism $G_2 \to \mathcal{U}(\mathcal{Q})$, then $\Theta\circ\varphi$ is a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$, so $\Theta(\varphi(x)) = 1$. For the second statement, Theorem A gives $w \in \mathrm{Res}_{\mathrm{MF}}(E)$. Hence $1 \neq \varphi(w) \in \mathrm{Res}_{\mathrm{MF}}(G_2)$. An injective homomorphism $G_2 \to \mathcal{U}(\mathcal{Q})$ would map $\varphi(w)$ to 1, a contradiction. $\square$

Proposition S9.3 (the universal MF quotient)**.** *Let G_1 be a countable group and $R = \mathrm{Res}_{\mathrm{MF}}(G_1)$.*

(1) R is the kernel of a single homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$.

(2) G_1/R is MF.

(3) Every homomorphism from G_1 to an MF group factors through G_1/R.

In particular G_1 is MF if and only if $R = \{1\}$.

Proof. (3) If $\varphi\colon G_1 \to M$ with M MF and $j\colon M \to \mathcal{U}(\mathcal{Q}((d_n)))$ is an injective homomorphism to $\mathcal{U}(\mathcal{Q})$, then $j \circ \varphi$ is a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$, so $j(\varphi(R)) = 1$ and, by injectivity of j, $\varphi(R) = 1$.

(1) and (2). Set $Q = G_1/R$. For every $\bar{g} \neq 1$ in Q, the definition of R gives a homomorphism $\Theta_{\bar{g}}\colon Q \to \mathcal{U}(\mathcal{Q})$ with $\Theta_{\bar{g}}(\bar{g}) \neq 1$. Enumerate the nonidentity elements $\bar{g}_1, \bar{g}_2, \ldots$ and choose coordinate unitary lifts of the corresponding homomorphisms. At stage k, choose in the ith lift, for each $i \leq k$, a coordinate at which multiplicativity holds within $1/k$ on the first k group elements and at which the designated element $\bar{g}_i$ stays separated by its fixed positive constant; such a coordinate exists because $\Theta_{\bar{g}_i}$ separates $\bar{g}_i$ along infinitely many coordinates. Take the block sum of the k chosen coordinates. Block operator norms are maxima, so the single block coming from $\Theta_{\bar{g}_i}$ already separates $\bar{g}_i$, and the sum therefore separates all of $\bar{g}_1, \ldots, \bar{g}_k$ at once. The resulting sequence is an operator-norm asymptotic representation separating every nonidentity element from the identity; its class, composed with the polar-correction isomorphism of Lemma 2.2, is a faithful homomorphism $Q \to \mathcal{U}(\mathcal{Q})$. This proves (2), and composing with $G_1 \twoheadrightarrow Q$ proves (1). Finally, if G_1 is MF then an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ gives $R = \{1\}$; if $R = \{1\}$ then (2) says G_1 is MF. $\square$

Corollary S9.4 (exact residual from a candidate quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \leq \mathrm{Res}_{\mathrm{MF}}(G_1)$. Every homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$ factors uniquely through G_1/N. If in addition G_1/N is MF, then*

$$\mathrm{Res}_{\mathrm{MF}}(G_1) = N.$$

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every such Θ. For the reverse containment in the displayed equality, compose the quotient map $G_1 \to G_1/N$ with an injective homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$: its kernel is exactly N, while the MF residual lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$. $\square$

Corollary S9.5 (residual reduction to the quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \leq \mathrm{Res}_{\mathrm{MF}}(G_1)$ and let $q\colon G_1 \to G_1/N$ be the quotient map. Then*

$$\mathrm{Res}_{\mathrm{MF}}(G_1) \;=\; q^{-1}(\mathrm{Res}_{\mathrm{MF}}(G_1/N)).$$

Proof. For the forward containment, Lemma S9.2 gives $q(\mathrm{Res}_{\mathrm{MF}}(G_1)) \subseteq \mathrm{Res}_{\mathrm{MF}}(G_1/N)$. Conversely, suppose $q(x) \in \mathrm{Res}_{\mathrm{MF}}(G_1/N)$ and let Θ be a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$. Since $N \leq \mathrm{Res}_{\mathrm{MF}}(G_1)$, Θ factors as $\bar{\Theta} \circ q$ with $\bar{\Theta}$ a homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$; hence $\Theta(x) = \bar{\Theta}(q(x)) = 1$. $\square$

Corollary S9.6. *Let A be a unital C^*-algebra admitting an injective, possibly nonunital, $*$-homomorphism into a matrix quotient. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.*

Proof. Let $j\colon A \to \mathcal{Q}$ be the injective, possibly nonunital, $*$-homomorphism and let $\phi\colon E \to \mathcal{U}(A)$. Define

$$\widehat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(\mathcal{Q}).$$

Then $\widehat{j}$ is an injective homomorphism on $\mathcal{U}(A)$, so $\widehat{j} \circ \phi$ is a homomorphism $E \to \mathcal{U}(\mathcal{Q})$. By Theorem A, $(\widehat{j} \circ \phi)(w) = 1$; injectivity of $\widehat{j}$ gives $\phi(w) = 1$. Proposition S2.3 has $w \neq 1$, so ϕ cannot be injective. $\square$

S9.1. **Permanence and its failure.**

Lemma S9.7 (permanence)**.** *Let all groups be countable.*

(1) Every subgroup of an MF group is MF.
(2) Every residually finite group is MF.
(3) Every locally finite group is MF.

Proof. (1) Restrict an MF embedding of the ambient group to the subgroup. (2) For a finite test set F, a finite quotient injective on F, composed with its left regular permutation action, is an exact finite-dimensional representation; distinct permutations differ in some column of a permutation matrix, giving operator-norm separation at least 1, and the local-to-global direction of Proposition 2.3 concludes. (3) A finite test set F lies in a finite subgroup K. The left regular representation $\lambda_K\colon K \to \mathcal{U}(\mathbb{C}^K)$ is a finite-dimensional unitary representation of K by permutation matrices. Because a local model is defined on all of G, set

$$V(g) = \lambda_K(g) \quad (g \in K), \qquad V(g) = 1 \quad (g \notin K).$$

On the test set $F \subseteq K$, this map is exactly multiplicative, since products of test elements lie in K and λ_K is a homomorphism; the values outside K do not enter the two conditions. It separates as in (2). $\square$

Corollary S9.8 (quotients)**.** *MF groups are not closed under quotients: the free group F_8 is MF, and the explicit group E is its quotient but is not MF.*

Proof. Map the eight free generators onto the eight displayed generators of E. The resulting homomorphism is surjective because those generators define the displayed presentation. The group F_8 is residually finite, hence MF by Lemma S9.7(2), while E is not MF by Theorem A. $\square$

S10. Orbit collapse

The argument of Theorem 4.1 extends from the negative spectral corner of a central involution to an arbitrary projection of $\mathcal{Q}$ with commuting conjugation orbit.

Theorem S10.1 (a commuting orbit forces commutation)**.** *Let H be countable, let $L \leq H$ have property* (T)*, let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism. Let $p \in \mathcal{Q}$ be a projection such that p commutes with $\Theta(s\gamma s^{-1})$ for every $\gamma \in L$ and the projections $\Theta(\gamma)\, p\, \Theta(\gamma)^*$, $\gamma \in L$, commute pairwise. Then p commutes with $\Theta(L)$.*

Proof. Property (T) makes L finitely generated; fix a finite symmetric generating set $S=\{a_1,\dots,a_m\}$. Write $p_g=\Theta(g)\,p\,\Theta(g)^*$ and $d_g=p_g-p$ for $g\in L$, and suppose $d_{g_0}\neq 0$ for some $g_0\in L$.

Since p_g and p are commuting projections, d_g is self-adjoint with $d_g^3=d_g$, so d_g^2 is a projection and $d_gd_g^2=d_g$; write $q_g=d_g^2$. All the p_g commute, hence so do all the d_g and all the q_g. Put

$$q=1-\prod_{a\in S}(1-q_a),$$

the join of the projections q_a, $a\in S$. The elements g with $p_g=p$ form a subgroup of L, so if $q=0$, and hence $d_a=0$ for every $a\in S$, then $d_g=0$ for every $g\in L$, contradicting $d_{g_0}\neq 0$. Thus $q\neq 0$.

The q-normalized Hilbert space. Take a self-adjoint bounded lift (x_n) of q. Then $\|x_n^2-x_n\|\to 0$, so the spectrum of x_n clusters at $\{0,1\}$ and continuous functional calculus makes $Q_n=\chi_{[1/2,\infty)}(x_n)$ a projection with $\|Q_n-x_n\|\to 0$; the sequence (Q_n) again lifts q. Since $q\neq 0$, $Q_n\neq 0$ for infinitely many n; discard the remaining coordinates, relabel by $\mathbb{N}$, and put $k_n=\operatorname{rank}Q_n\geq 1$.

Give $M_{d_n}(\mathbb{C})$ the inner product $\langle x,y\rangle_n=\operatorname{Tr}(y^*x)/k_n$, write $\|x\|_{\mathrm{F}}=\operatorname{Tr}(x^*x)^{1/2}$ for the unnormalized Hilbert–Schmidt norm, and let K_ω be the Hilbert-space ultraproduct of these spaces along a free ultrafilter ω. Let $\mathcal{I}_q$ be the set of finite sums $\sum_{j\leq r}x_jqy_j$ with $x_j,y_j\in\mathcal{Q}$, a two-sided ideal of $\mathcal{Q}$ invariant under inner automorphisms.

For $z=\sum_{j\leq r}x_jqy_j\in\mathcal{I}_q$, choose bounded coordinate lifts $X_{j,n}$ of x_j and $Y_{j,n}$ of y_j and put $Z_n=\sum_{j\leq r}X_{j,n}Q_nY_{j,n}$; then $\operatorname{rank}(Z_n)\leq rk_n$. The square of the Frobenius norm is the sum of the squares of the singular values; at most $\operatorname{rank}(Z_n)$ of them are nonzero and each is at most the operator norm, so

$$\|Z_n\|_{\mathrm{F}}^2\ \leq\ \operatorname{rank}(Z_n)\,\|Z_n\|^2\ \leq\ r\,k_n\,\|Z_n\|^2 \tag{10}$$

and (Z_n) is a bounded sequence for the inner products $\langle\cdot,\cdot\rangle_n$. Its class in K_ω depends only on $z=\sum_j x_jqy_j$: a sequence Z_n' obtained from a second decomposition, with r' summands, satisfies $\|Z_n-Z_n'\|\to 0$ and $\operatorname{rank}(Z_n-Z_n')\leq(r+r')k_n$, so $\|Z_n-Z_n'\|_{\mathrm{F}}^2/k_n\to 0$ by (10). Write $\Lambda(z)=[Z_n]_\omega$. The map $\Lambda\colon\mathcal{I}_q\to K_\omega$ is linear,

$$\|\Lambda(q)\|=\lim_\omega\Big(\frac{\operatorname{Tr}Q_n}{k_n}\Big)^{1/2}=1,$$

and $\Lambda(z)=0$ implies $\Lambda(xzy)=0$ for all $x,y\in\mathcal{Q}$, since bounded lifts of x and y multiply $\|Z_n\|_{\mathrm{F}}$ by a bounded factor.

An exact cocycle. For $a\in S$ the projection q_a annihilates $\prod_{b\in S}(1-q_b)$, so $q_aq=q_a$ and $d_a=d_aq_a=d_aq\in\mathcal{I}_q$. From $p_{gh}=\Theta(g)\,p_h\,\Theta(g)^*$,

$$d_{gh}=d_g+\Theta(g)\,d_h\,\Theta(g)^*\qquad(g,h\in L), \tag{11}$$

and $\mathcal{I}_q$ is inner-invariant, so $d_g\in\mathcal{I}_q$ for every $g\in L$.

Choose unitary coordinate lifts $U_n(h)$ of $\Theta(h)$ (Lemma 2.2). Conjugation by a unitary preserves each $\langle\cdot,\cdot\rangle_n$, and $\|\mathrm{Ad}\,u - \mathrm{Ad}\,v\| \le 2\,\|u-v\|$, so $\pi(h) = [\mathrm{Ad}\,U_n(h)]_\omega$ defines a unitary representation of H on K_ω, and $\pi(h)\Lambda(z) = \Lambda(\Theta(h)\,z\,\Theta(h)^*)$ on the closed invariant subspace $K_q = \overline{\Lambda(\mathcal{I}_q)}$. By (11) the map $\beta(g) = \Lambda(d_g)$ satisfies

$$\beta(gh) = \beta(g) + \pi(g)\beta(h) \qquad (g,h \in L),$$

a 1-cocycle for $\pi|_L$ with values in K_q.

Suppose $\beta(a) = 0$ for every $a \in S$. Put $r_1 = 1$ and $r_i = \prod_{j<i}(1-q_{a_j})$ for $i \ge 2$. The products $e_i = r_i q_{a_i}$ are projections with $e_i = r_i - r_{i+1}$, so they are pairwise orthogonal and $\sum_i e_i = 1 - r_{m+1} = q$. Each d_{a_i} commutes with r_i, so $e_i = (d_{a_i} r_i)\,d_{a_i}$ and $\Lambda(e_i) = 0$, whence $\Lambda(q) = 0$ against $\|\Lambda(q)\| = 1$. So β is not identically zero.

Property (T). By the Delorme–Guichardet theorem [BHV, Theorem 2.12.4], β is a coboundary: there is $y \in K_q$ with $\beta(g) = y - \pi(g)y$ for every $g \in L$. The hypothesis on p gives $d_{sas^{-1}} = 0$, hence $\pi(sas^{-1})y = y$, for every $a \in L$. The fixed-space comparison gives $\mathrm{Fix}\,\pi(sLs^{-1}) = \mathrm{Fix}\,\pi(L)$: the projection P onto $\mathrm{Fix}\,\pi(L)$ and its conjugate $Q = \pi(s)P\pi(s)^*$, whose range is $\mathrm{Fix}\,\pi(sLs^{-1})$, satisfy $P \le Q$ and $P \sim Q$ in the norm ultraproduct, which is finite, so $P = Q$ by Lemma 3.1. Theorem S4.1(1) gives the same projection comparison with the weight k_n in place of d_n. Hence $\pi(g)y = y$ and $\beta(g) = 0$ for every $g \in L$, contradicting the previous paragraph. So $d_g = 0$, that is, p commutes with $\Theta(g)$, for every $g \in L$. □

Equivalently, every projection of $\Theta(sLs^{-1})' \cap \mathcal{Q}$ whose $\Theta(L)$-conjugates commute pairwise already lies in $\Theta(L)' \cap \mathcal{Q}$.

Definition S10.2 (involutive orbit witness)**.** Let H be a group, $L \le H$ a subgroup, and $s \in H$. An element $k \in H$ is an *involutive orbit witness* for (L,s) if

(W1) $k^2 = 1$;
(W2) $s\gamma s^{-1}$ commutes with k for every $\gamma \in L$;
(W3) the conjugates $\gamma_1 k \gamma_1^{-1}$ and $\gamma_2 k \gamma_2^{-1}$ commute for all $\gamma_1, \gamma_2 \in L$.

The pair (L,s) determines the normal subgroup

$$D_{\mathrm{coll}}(L,s) \;=\; \big\langle\!\big\langle\, [\gamma,k] \;:\; \gamma \in L,\ k \text{ a witness for } (L,s) \,\big\rangle\!\big\rangle_H.$$

In a permutational wreath product $K^{(X)} \rtimes G$ with $L \le G$ conjugated into itself by s and $x_0 \in X$ a point fixed by sLs^{-1}, every involution in the copy of K supported at x_0 is a witness: distinct coordinate copies commute, and sLs^{-1} fixes x_0.

Theorem S10.3 (the orbit of an involutive witness)**.** *Let H be countable, let $L \le H$ be a subgroup with property* (T)*, and let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$. Then*

$$D_{\mathrm{coll}}(L,s) \;\le\; \mathrm{Res}_{\mathrm{MF}}(H).$$

Proof. Let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism and k a witness for (L, s). By (W1) the unitary $u = \Theta(k)$ is self-adjoint, so $p = \frac{1}{2}(1-u)$ is a projection, and

$$\Theta(\gamma)\, p\, \Theta(\gamma)^* = \tfrac{1}{2}(1 - \Theta(\gamma k \gamma^{-1})) \qquad (\gamma \in L).$$

By (W2) the projection p commutes with $\Theta(s\gamma s^{-1})$ for $\gamma \in L$, and by (W3) the displayed projections commute pairwise. By Theorem S10.1, p commutes with $\Theta(L)$; hence so does $u = 1 - 2p$, and $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. Kernels are normal, so Θ is trivial on $D_{\mathrm{coll}}(L, s)$. □

Theorem S10.4 (the orbit of a finite-order witness)**.** *Let H be countable, let $L \le H$ have property* (T)*, let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $k \in H$ have finite order. If the conjugate copy sLs^{-1} centralizes k and the conjugates $\gamma k \gamma^{-1}$, $\gamma \in L$, commute pairwise, then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$; the normal closure of all such commutators, over all finite-order witnesses for (L, s), lies in* $\mathrm{Res}_{\mathrm{MF}}(H)$*, with equality whenever the corresponding quotient of H is MF.*

Proof. Let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$, m the order of k, and $u = \Theta(k)$, a unitary with $u^m = 1$. Its discrete Fourier idempotents $p_j = m^{-1} \sum_{t \in \mathbf{Z}/m} e^{-2\pi i j t/m} u^t$, for $j \in \mathbf{Z}/m$, are self-adjoint idempotents of $\mathcal{Q}$. Each p_j is a polynomial in u, so every element commuting with u commutes with every p_j. Thus $\Theta(sLs^{-1})$ centralizes each p_j, and the $\Theta(L)$-conjugates of each p_j commute pairwise because the conjugated witnesses do. By Theorem S10.1, every p_j commutes with $\Theta(L)$, and Fourier reconstruction expresses $u = \sum_j e^{2\pi i j/m} p_j$, so u commutes with $\Theta(L)$ and $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. □

Proof of Theorem B. Write $q\colon E \to E/\langle w \rangle$, $\bar{d} = q(d)$, and $\bar{L} = q(\iota(\mathcal{B}))$. The element $\bar{d}$ is an involution because $c^2 = 1$, and $q(t)\bar{L}q(t)^{-1}$ centralizes $\bar{d}$ because $t\iota(\mathcal{B})t^{-1}$ commutes with c in E. Property (T) passes to $\bar{L}$, and $q(t)$ conjugates it into itself. Thus (W1) and (W2) of Definition S10.2 hold, and only (W3) remains.

The linear generators x, y, z commute with d, and t carries each v_i to v_i^2, so $\iota(v_i^2)$ commutes with d as well. Hence $\iota(g)\, d\, \iota(g)^{-1}$ depends only on the exponent vector of the translation part of g modulo 2. Hence there are at most eight such conjugates, indexed by $(\mathbb{Z}/2)^3$. By Theorem 5.1 the distinguished word is $w = [d, \iota(v_1)\, d\, \iota(v_1)^{-1}]$, the commutator attached to the class of v_1. The six words x, x^2, z, yz, yx, y^2x in the linear generators send the class of v_1 to the six remaining nonzero classes, and each commutes with d. Conjugating the displayed identity by the corresponding $\iota(r)$ shows that every nonzero parity commutator is a conjugate of w; the zero class gives $[d, d] = 1$. Since w is central, all eight lie in $\langle w \rangle$. Conjugating a commutator of two arbitrary conjugates by the inverse of one of them reduces it to this case. Thus $\bar{d}$ is an involutive orbit witness for $(\bar{L}, q(t))$, and by Theorem S10.3 $[q(\iota(v_1)), \bar{d}] \in \mathrm{Res}_{\mathrm{MF}}(E/\langle w \rangle)$.

The homomorphism $E \to W$ of Proposition S2.3 sends $[\iota(v_1), d]$ to the product of the Clifford generators at the two distinct points τo and $j(\bar{v}_1)\tau o$, and sends $\langle w \rangle$ into $\langle \zeta \rangle$. The product of the Clifford generators at two distinct points does not lie in $\langle \zeta \rangle$, so $[\iota(v_1), d] \notin \langle w \rangle$ and its image in $E/\langle w \rangle$ is nontrivial. By Proposition S9.3, this nontrivial element of the MF residual makes the quotient non-MF. Corollary S9.5 identifies $\mathrm{Res}_{\mathrm{MF}}(E)$ with the q-preimage of $\mathrm{Res}_{\mathrm{MF}}(E/\langle w \rangle)$, so u belongs to it. □

Corollary S10.5 (collapse computes the residual)**.** *In the setting of Theorem S10.3, write $q\colon H \to H/D_{\mathrm{coll}}(L, s)$ for the quotient map. Then*

$$\mathrm{Res}_{\mathrm{MF}}(H) \;=\; q^{-1}\big(\mathrm{Res}_{\mathrm{MF}}(H/D_{\mathrm{coll}}(L, s))\big),$$

and if $H/D_{\mathrm{coll}}(L, s)$ is MF, then $\mathrm{Res}_{\mathrm{MF}}(H) = D_{\mathrm{coll}}(L, s)$.

Proof. Combine Theorem S10.3 with Corollaries S9.5 and S9.4. □

If $H/D_{\mathrm{coll}}(L, s)$ is MF, then it is the universal MF quotient of H (Proposition S9.3): every homomorphism from H to an MF group factors through it.

S11. Group algebras, soficity, and permanence

S11.1. **The reduced group algebra of E.** The stable finiteness in Theorem C comes from two standard facts.

Lemma S11.1. *Let G_1 be a countable discrete group.*

(1) The canonical tracial state τ on $C^*_{\mathrm{r}}(G_1)$ is faithful.

(2) Every unital C^*-algebra A with a faithful tracial state is stably finite: for every $k \geq 1$, every isometry in $M_k(A)$ is a unitary.

Proof. (1) Realize $C^*_{\mathrm{r}}(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g. Let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$. Each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C^*_{\mathrm{r}}(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}} x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k}\sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k}\sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i-th diagonal entry of x^*x is $\sum_j x^*_{ji}x_{ji}$, so $\tau_k(x^*x) = 0$ implies $\tau(x^*_{ji}x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j. If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. □

S11.2. **The Clifford quotient and permanence.** The map $E \to W$ of Proposition S2.3 is surjective: its image contains Σ and the generator c_o at the base point, hence every Clifford generator by transitivity of the Σ-action on X. Since E is generated by v_1, x, y, z, t, c (Section S1), their six images generate W. The map sends $w \in \mathrm{Res}_{\mathrm{MF}}(E)$ to the nontrivial sign ζ, so

$\zeta \in \mathrm{Res}_{\mathrm{MF}}(W)$ by Lemma S9.2. Hence the finitely generated group W is not MF. Neither is $C^*_{\mathrm{r}}(W)$: if $C^*_{\mathrm{r}}(W)$ embedded in a matrix quotient, the map $u \mapsto e(u) + 1 - e(1)$ would give an injective homomorphism $W \to \mathcal{U}(\mathcal{Q})$, as in the proof of Theorem A.

Failure of extension permanence. MF groups are not closed under extensions. More precisely, W fits into an exact sequence

$$1 \longrightarrow \mathrm{Cl}(X) \longrightarrow W \longrightarrow \Sigma \longrightarrow 1$$

whose kernel is locally finite — hence LEF, sofic and MF — and whose quotient is MF, while W itself is finitely generated and not MF. For comparison, an extension of a residually finite group by a residually finite group is weakly sofic [Gl], and weak soficity is detected by finite targets [GR].

Proof. By Lemma S9.7(3), the Clifford group is locally finite, hence MF. The group Σ identifies with the subgroup of $\mathrm{GL}_4(\mathbb{Q})$ generated by $\bar{\Gamma}$ and D: the levels are $D^{-n}\bar{\Gamma}D^n$, and the determinant gives the $\mathbb{Z}$-coordinate. It is residually finite. Modulo every odd integer, the dilation matrix is invertible because its determinant is 8; hence the congruence reduction of the integral affine model is compatible with the dilation and extends over the direct limit and its shift. These finite quotients separate every nontrivial element of Σ, consistent with Mal'cev's theorem for finitely generated linear groups [Mal]. Thus Σ is MF by Lemma S9.7(2), whereas W is not MF by the preceding construction. □

Extensions by $\mathbb{Z}$. Composing the two canonical projections gives a surjection $W \to \mathbb{Z}$ whose kernel is $\mathrm{Cl}(X) \rtimes T$, and this kernel is LEF, hence MF: every finite subset lies in $L \rtimes \bar{\Gamma}_n$ for a finite invariant subgroup $L \leq \mathrm{Cl}(X)$ and one level, and that group is residually finite because the level acts on the finite group L through a finite quotient. Thus W is an extension of an LEF group by $\mathbb{Z}$ and is not MF, while soficity is preserved by every extension with amenable quotient, as in Elek–Szabó [ES06].

A simple sofic envelope. There is a countable simple sofic group S_{simp} with

$$\mathrm{Res}_{\mathrm{MF}}(S_{\mathrm{simp}}) = S_{\mathrm{simp}}.$$

Equivalently, every homomorphism from S_{simp} to an MF group is trivial.

Proof. By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed E, which is sofic by Theorem D, in such a group. By functoriality of the MF residual, the image of w is a nontrivial element of the MF residual of the simple group. This residual is a nontrivial normal subgroup, so simplicity makes it the whole group. Every homomorphism to an MF group maps the MF residual to 1. □

S11.3. **Soficity of the literal group.** Deleting the letter c and the families (C) and (W) from (8) leaves the presentation $\langle\, \mathcal{B}, t \mid tut^{-1} = \bar{\alpha}(u)\,\rangle$ of the ascending HNN extension of the base along the doubling endomorphism. By Remark S1.2 the base is $\bar{\Gamma}$, so

$$E/\langle\langle c\rangle\rangle \;\cong\; \Sigma$$

for the group $\Sigma = T \rtimes \mathbb{Z}$ of Section S2, with its site set $X = \Sigma/B$, base point o, shift τ and level-zero base $B = j(\bar{\Gamma})$. The levels

$$\bar{\Gamma}_n = \tau^{-n} B \tau^n \qquad (n \geq 0)$$

form an ascending chain of copies of $\bar{\Gamma}$ with union T. Put $K = \tau^{-1}B\tau$; then $B \leq K$ and $[K : B] = [\bar{\Gamma} : \bar{\alpha}(\bar{\Gamma})]$, at most eight. The cosets Σ/K are the *blocks*, and each block, as a set of sites, has $[K : B]$ elements.

Let $\mathcal{G}$ be the graph on X whose edges are the Σ-orbit of the pair $\{\tau o,\, j(\bar{v}_1)\tau o\}$, whose entries are distinct by Lemma S2.2. Two sites are adjacent in $\mathcal{G}$ exactly when they are distinct and lie in a common block. Adjacent sites share a block because $(\tau o)^{-1}(j(\bar{v}_1)\tau o) = \tau^{-1} j(\bar{v}_1)\tau \in K$. Conversely, the chart $u\,\bar{\alpha}(\bar{\Gamma}) \mapsto \tau^{-1}j(u)\tau o$ identifies the block of o with $\bar{\Gamma}/\bar{\alpha}(\bar{\Gamma})$ and the generating edge with the pair $\{\bar{\alpha}(\bar{\Gamma}),\, \bar{v}_1\bar{\alpha}(\bar{\Gamma})\}$; since $\bar{\alpha}(\bar{\Gamma})$ acts transitively on the nontrivial cosets of $\bar{\alpha}(\bar{\Gamma})$, every pair of distinct sites of a block is the image of the generating pair.

For a graph $\mathcal{G}$ on X, let $C(\mathcal{G})$ be the group presented by generators ζ and c_ξ $(\xi \in X)$ and relations

$$\zeta^2 = c_\xi^2 = 1, \qquad [\zeta, c_\xi] = 1, \qquad [c_\xi, c_\eta] = \zeta \;\text{ for each edge } \{\xi, \eta\} \text{ of } \mathcal{G}.$$

For the complete graph on X this is $\mathrm{Cl}(X)$ of Construction S2.1, and adjoining the anticommutation relations at the remaining pairs gives a surjection $C(\mathcal{G}) \to \mathrm{Cl}(X)$.

Proposition S11.2 (normal form for E)**.** *There is an isomorphism*

$$E \;\cong\; C(\mathcal{G}) \rtimes \Sigma$$

under which $\langle\langle c\rangle\rangle$ is the factor $C(\mathcal{G})$, the conjugate gcg^{-1} is the lamp c_{gB}, and the distinguished word w is the sign ζ. Each level $\bar{\Gamma}_n$ has finite orbits on X.

Proof. Send the six base letters and t to the corresponding generators of Σ, and c to c_o. Families (B) and (T) of (8) hold in Σ by construction. For family (C): $c_o^2 = 1$, and B fixes o, so each base generator fixes c_o. For family (W): the pair $\{\tau o,\, j(\bar{v}_1)\tau o\}$ is an edge, so the image of w is $[c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = \zeta$, which is central in $C(\mathcal{G})$ and fixed by Σ. This defines $E \to C(\mathcal{G}) \rtimes \Sigma$.

In the other direction, $\zeta \mapsto w$ and $c_{gB} \mapsto gcg^{-1}$ respects the defining relations of $C(\mathcal{G})$: the assignment is independent of the coset representative because the base centralizes c; the images are involutions; w is central of order two; and along an edge $g\{\tau o,\, j(\bar{v}_1)\tau o\}$ the commutator of the two images is $gwg^{-1} = w$; the presentation imposes relations only along edges. Combined with the splitting $\Sigma \to E$ on the seven remaining letters, this gives $C(\mathcal{G}) \rtimes \Sigma \to E$. The two composites fix the eight letters of E and the generators ζ, c_ξ of $C(\mathcal{G})$, hence are the identity.

For the orbits, $\bar\alpha(\bar\Gamma)$ has index at most eight in $\bar\Gamma$ — the parity classes of the translation lattice — so τ conjugates B onto a subgroup of finite index in itself. Every $\bar\Gamma_n$ is therefore commensurable with B, all of Σ commensurates B, and orbit–stabilizer makes each $\bar\Gamma_n$-orbit in $X = \Sigma/B$ finite. □

Composing the isomorphism with $C(\mathcal{G}) \to \mathrm{Cl}(X)$ gives the surjection $E \to W$ of Proposition S2.3. The group $\mathrm{Cl}(X)$ is locally finite; $C(\mathcal{G})$ is not, since two lamps at sites of different blocks generate an infinite dihedral group and three blocks give a nonabelian free subgroup. For a finite set S of sites write

$$M_S = \langle\, \zeta,\ c_\xi\ (\xi \in S)\,\rangle \ \le\ C(\mathcal{G}).$$

Lemma S11.3 (finite-site subgroups are residually finite)**.** *For every finite set S of sites the subgroup M_S is residually finite.*

Proof. The quotient map by $\langle\zeta\rangle$ carries $C(\mathcal{G})$ onto the free product of the elementary abelian 2-groups of the blocks, and M_S into the free product of the finitely many factors at blocks meeting S. A free product of finite groups is residually finite. The kernel of $C(\mathcal{G}) \to C(\mathcal{G})/\langle\!\langle\zeta\rangle\!\rangle$ is the normal closure of ζ, and ζ is central, so that normal closure is $\langle\zeta\rangle$; hence the kernel of this map on M_S lies in $\langle\zeta\rangle$. The surjection $C(\mathcal{G}) \to \mathrm{Cl}(X)$ sends ζ to the sign of $\mathrm{Cl}(X)$, which is nontrivial, and $\mathrm{Cl}(X)$ is locally finite, so the image of the finitely generated subgroup M_S in $\mathrm{Cl}(X)$ is finite. An element of M_S in the kernel of the first map lies in $\langle\zeta\rangle$, and the second map separates the two elements of that subgroup. □

Proof of Theorem D. Let E_0 be the kernel of the homomorphism $E \to \mathbb{Z}$ given by the exponent sum in t. By Proposition S11.2, $E_0 \cong C(\mathcal{G}) \rtimes T$.

Take a finite subset of E_0. Its T-coordinates lie in one level $\bar\Gamma_n$, the chain being ascending, and its $C(\mathcal{G})$-coordinates involve only finitely many lamp generators and possibly the sign, so they lie in M_{S_0} for some finite set S_0 of sites. Let S be the union of the $\bar\Gamma_n$-orbits of the sites in S_0, a finite set by Proposition S11.2. Then M_S is $\bar\Gamma_n$-invariant and contains M_{S_0}, so the finite subset lies in $M_S \rtimes \bar\Gamma_n$.

That group is residually finite. Indeed M_S is residually finite by Lemma S11.3, the level $\bar\Gamma_n \cong \bar\Gamma$ is residually finite through the congruence quotients of its matrix model (Lemma S2.2), and the action of $\bar\Gamma_n$ on M_S factors through a finite permutation group F because S is finite. The kernel of $\bar\Gamma_n \to F$ has finite index, and the corresponding finite-index subgroup of the semidirect product is $M_S \times \ker(\bar\Gamma_n \to F)$, which is residually finite; residual finiteness passes to a group containing a residually finite subgroup of finite index.

Hence E_0 is a directed union of subgroups that embed in residually finite groups; that is, E_0 is LEF and therefore sofic. The group E is a split extension of E_0 by $\mathbb{Z}$, split by t, so the Elek–Szabó amenable-extension theorem implies that E is sofic [ES06], and hence hyperlinear. By Theorem A, E is not MF. LEF groups are MF [CDE], so E is not LEF; a residually finite group is LEF, so E is not residually finite either. $\square$

The same decomposition also proves that E is exact. The quotient of $C(\mathcal{G})$ by $\langle\zeta\rangle$ is the free product of the elementary abelian 2-groups of the blocks, each of order at most 2^8, and free products, directed unions and extensions of exact groups are exact [KW2]; so that free product is exact, and $C(\mathcal{G})$, a central extension of it by $\langle\zeta\rangle$, is exact as well. The group Σ is a subgroup of $\mathrm{GL}_4(\mathbb{Q})$ and therefore exact [GHW], and $E \cong C(\mathcal{G}) \rtimes \Sigma$. For a discrete group exactness is equivalent to exactness of the reduced group algebra [KW1], so $C^*_{\mathrm{r}}(E)$ is exact. The same applies to W, whose Clifford kernel is locally finite and hence amenable.

S12. Marked limits, quasi-identities, and undecidability

S12.1. **Marked-limit closure.**

Theorem S12.1 (the MF locus is closed)**.** *Fix $k \geq 1$ and consider the space of k-marked groups — quotients of the free group F_k, with the usual topology in which a basic clopen set specifies the truth values of finitely many word equations in the generators.*

(1) The marked groups with MF quotient form a closed set.
(2) Equivalently, the non-MF marked groups form an open set.
(3) Every non-MF k-marked group has a finite-radius obstruction: there is a radius R such that every k-marked group whose relations agree with it on the reduced words of length at most R is non-MF.

Proof. A marked group fails MF exactly when normalized local approximation fails on some finite subset of it. Choose words representing the finitely many elements and products involved in such a failed local approximation. The required equalities and inequalities depend on finitely many word relations, so the same failure holds throughout a basic clopen neighborhood. Taking R at least the lengths of these finitely many words gives the finite-radius statement. $\square$

S12.2. **A finite universal Horn obstruction.** The presentation of E yields a finite quasi-identity – equivalently, a universal Horn sentence – satisfied by every MF group. Write R for the forty-one relators of (8), regarded as words in eight variables, and w for the distinguished word in those variables.

Proposition S12.2 (finite quasi-identity)**.** *Every MF group G satisfies the finite universal Horn sentence*

$$\forall x_1, \ldots, x_8\colon \quad \bigwedge_{r\in R} r(x_1,\ldots,x_8) = 1 \implies w(x_1,\ldots,x_8) = 1,$$

while the canonical generating tuple of E satisfies every relator in R and has $w \neq 1$.

Proof. A tuple satisfying the relators induces a homomorphism $E \to G$ sending the distinguished word of E to $w(x_1, \ldots, x_8)$. If G is MF, the image is trivial, because $w \in \mathrm{Res}_{\mathrm{MF}}(E)$ by Theorem A and the residual is functorial (Lemma S9.2). The canonical tuple of E satisfies R by definition, and its distinguished word is nontrivial by Theorem A. □

Corollary S12.3 (a basic clopen set of non-MF marked groups)**.** *In the space of eight-generator marked groups — quotients of the free group F_8, topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of R and $w \neq 1$ form a nonempty clopen set consisting entirely of non-MF groups.*

Proof. Both conditions depend on finitely many word equalities and inequalities, so the set is clopen. It contains the marked group E by Theorem A, and Proposition S12.2 shows that none of its points is MF. □

S12.3. **Undecidability of MF recognition.** Lemma S9.7(1) and Theorem A show that MF is a Markov property of finitely presented groups: the trivial group is MF, while by subgroup heredity the finitely presented group E embeds in no MF group. The Adian–Rabin theorem therefore implies that MF recognition is undecidable [Ra].

Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_e for the group presented by the code e.

Corollary S12.4 (undecidability of MF recognition)**.** *Write W for the predicate on pairs consisting of a presentation code and a word in the generators of that code, holding when the word is trivial in the presented group. Then W is undecidable, and no algorithm decides, from a presentation code, whether the presented group is MF. Moreover W is recursively enumerable, so its complement is not, and neither is the set of presentation codes of non-MF groups.*

Proof. For the undecidability of W, take a modular machine whose configuration halting problem is undecidable and the finitely presented group built from it. That group has a presentation code; the halting word of a configuration is a computable function of the configuration in the numbering that code uses; and the code's word problem holds on that word exactly when the configuration halts. A decision procedure for W would compose with that map to decide halting.

For the reduction, take the free product of the input presentation with a fixed presentation of E, and apply the Adian–Rabin construction to the input word. The resulting presentation code is computable from the pair and presents an MF group exactly when the input word is trivial: in the trivial case the output group is the trivial group, and in the nontrivial case the output group contains E and is non-MF by Lemma S9.7(1). Composing a decision procedure for MF recognition with this map would therefore decide W, so no such procedure exists. For the last assertion, the positive instances of W are exactly those carrying a finite certificate — a list of conjugators and relators, written as raw data — and whether a certificate works is decided by a search over raw words, so W is recursively enumerable. Were its complement recursively enumerable as well, W would be decidable, which it is not. Finally, the preimage of the set of non-MF presentation codes under this computable map is exactly the set of pairs outside W, and the preimage of a recursively enumerable set under a computable map is recursively enumerable. $\square$

The reduction requires an effective coding of presentations and a computable transformation of codes; without these conditions it would not establish undecidability.

The source of the undecidability is proved here as well. Undecidability of W is classical, and a finitely presented group carrying a sequence of words whose triviality is undecidable is available here, with no hypothesis and no literature input, by way of Aanderaa–Cohen modular machines instead of Boone–Britton. Matching that group to W is a question of coordinates rather than of group theory — W is a predicate on codes, while the existence statement supplies a group and a sequence of words — and the two sides are now matched, in four steps. The words are written as raw data in the numbering the code uses, so no transport along a coding is needed to read them. The map sending a configuration to its word is primitive recursive, hence computable, even though the generator indices it names are obtained by choice: for a fixed presentation those indices are fixed natural numbers, and the word is built from them by repetition and concatenation, so the map of the varying argument is computable whether or not the indices can be evaluated. The code's word problem on that word is triviality of the halting element, a renumbering away from what the presentation already proves; and the halting element is trivial exactly when the configuration halts, which is what the group was built for. Agreement with halting is therefore a theorem and not an assumption. Joining these at the group of a machine whose configuration halting problem is undecidable leaves W undecidable outright, and the corollary carries no hypothesis.

The negative side needs one more ingredient, and it is the same coordinate discipline. A certificate for $w = 1$ is naturally a list of conjugators and relators in the free group on the code's generators, but that free group depends on the code, so no search can range over it; written instead as raw words, and with the equality test replaced by a deletion sequence witnessing free triviality, the certificates become data a single unbounded search can enumerate. That makes W recursively enumerable, and an undecidable recursively enumerable predicate has a complement that is not.

There are infinitely many pairwise nonisomorphic finitely presented non-MF groups, and continuum many pairwise nonisomorphic finitely generated non-MF groups. The groups $E \times \mathbb{Z}^k$ for $k \geq 0$ are finitely presented, pairwise nonisomorphic, and non-MF, each containing E (Lemma S9.7(1)). They are pairwise nonisomorphic because their abelianizations have distinct torsion-free ranks; the number of homomorphisms to $\mathbb{Z}/2$ separates them as well. For finitely generated examples, choose continuum many pairwise nonisomorphic finitely generated groups N, for instance Neumann's two-generator groups [Ne]. Each product $E \times N$ is non-MF. A fixed countable group X can be isomorphic to $E \times N$ for at most countably many isomorphism types of N, because each such N embeds in X as a finitely generated subgroup and X has only countably many finitely generated subgroups. Hence the products yield $2^{\aleph_0}$ pairwise nonisomorphic finitely generated non-MF groups.

S13. Limitations of the operator-norm method

The proof is specific to the operator norm. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm: the estimate $\|\mathrm{Ad}(U) - \mathrm{Ad}(V)\| \leq 2\,\|U - V\|$ has no converse, and there are unitaries of arbitrarily small normalized Hilbert–Schmidt distance whose adjoints are at operator-norm distance 2. Operator-norm control is needed here to ensure that the classes $[\mathrm{Ad}\, U_n(g)]_\omega$ are multiplicative; Hilbert–Schmidt control alone does not provide this.

Adding an identity block leaves operator-norm multiplicative defects and separations unchanged but decreases normalized Hilbert–Schmidt distances. By taking the identity block sufficiently large, all pairwise normalized Hilbert–Schmidt distances can be made smaller than any prescribed positive bound. Thus 2-norm triviality of an element does not by itself prove a group non-MF. The central-involution argument instead passes to an invariant corner and normalizes the trace on that corner, so the 2-norm separation is independent of the corner's relative rank: the corner $q_n M_{d_n}(\mathbb{C}) q_n$ is untouched by padding while d_n grows, and r_n/d_n may tend to zero without affecting the contradiction. The spectral projection of Theorem S5.6 is normalized on its own range in the same way, and the proof of Theorem S10.1 normalizes by the rank k_n.

Stable finiteness and faithful traces on the coordinate algebras do not suffice for Theorem 3.4. For a faithfully traced unital C^*-algebra A, the left and right actions on $L^2(A, \tau)$ can generate an infinite von Neumann algebra; for an infinite-dimensional II_1 factor in standard form they generate all of $B(L^2(A, \tau))$. Constant coordinates $A_n = C^*_{\mathrm{r}}(E)$ give an explicit counterexample: the left regular representation embeds E exactly and injectively into the norm quotient

$$\textstyle\prod_n A_n / \bigoplus_n A_n,$$

the element c centralizes the Kazhdan base there, and conjugation by t produces tct^{-1}, whose commutator with the base is the element u with $u^2 = w \neq 1$ of Theorem A. Hence Theorem 3.4 fails when its matrix coordinates are replaced by stably finite unital C^*-algebras carrying faithful tracial states and its normalized Hilbert–Schmidt norm by the trace 2-norm, with all its remaining hypotheses unchanged.

Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]. Their question concerns Hamming and tracial models, whereas the adjoint spectral estimates here require operator-norm multiplicative control, which permutation and Hilbert–Schmidt models do not supply. The soficity of E is proved separately, from its normal form (Theorem D).

Questions.

(1) What is $\mathrm{Res}_{\mathrm{MF}}(E)$? It properly contains $\{1, w\}$ by Theorem B, and Corollary S10.5, applied to $E/\langle w\rangle$ and its defect $D = D_{\mathrm{coll}}$, computes it from $\mathrm{Res}_{\mathrm{MF}}((E/\langle w\rangle)/D)$. These reductions give no further information about that last group, because the elements they detect have already been killed: w is trivial in $E/\langle w\rangle$, and the commutators generating D are trivial in the quotient by D. That quotient may still contain a finite normal subgroup, a normal Kazhdan subgroup, or an orbit configuration to which the preceding criteria apply.

(2) Is there a torsion-free finitely presented non-MF group? For torsion-free groups, Theorem S5.3 gives no obstruction because every finite subgroup is trivial, while Theorem S5.6 does apply. What remains is group-theoretic: a torsion-free finitely presented group whose subgroup N_{conj} contains a *nontrivial* normal property-(T) subgroup. Remark S13.1 gives a small-cancellation route over the torsion-free finitely presented Kazhdan group of [FFF], which removed the analogous torsion obstacle for nonsoficity. The Clifford construction of Theorem 5.1 cannot produce such a group: it imposes $c^2 = 1$, so $d^2 = 1$, and a torsion-free quotient q then has $q(d) = 1$ and $q(N_{\mathrm{conj}}) = 1$.

(3) Does MF imply hyperlinearity? An MF embedding controls operator norm but does not require the induced tracial model to separate nontrivial group elements, so it does not directly produce a hyperlinear model. This question concerns the CDE convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.

(4) For which groups do the criteria of Theorems S5.3, S5.6 and S10.1 compute the residual exactly?

(5) Does the obstruction survive when the matrix blocks are replaced by other building blocks? The proof of Theorem 3.4 uses finite dimensionality of $M_{d_n}(\mathbb{C})$ twice: for the conjugation action on $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$, and for finiteness of the norm ultraproduct. Faithful traces alone do not replace it. For which separable C^*-algebras A_n does every homomorphism $H \to \mathcal{U}(\prod_n A_n / \bigoplus_n A_n)$ still map w to the identity?

Unitary lifting itself works for arbitrary unital coordinate algebras: if u is unitary in the quotient and (x_n) is any lift, then $\|x_n^*x_n - 1\| \to 0$ and $\|x_nx_n^* - 1\| \to 0$, so x_n is invertible for all large n and the polar correction $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \to 0$, with no real-rank-zero or semiprojectivity hypothesis in the sense of Loring [Lo98]. The matrix-specific inputs are the Hilbert-space conjugation model and finiteness of the norm ultraproduct, not the lifting step.

Remark S13.1 (a small-cancellation realization). Fournier-Facio has pointed out (personal communication, 2026) that the affine base of Sections S1–S2 can be replaced by a small-cancellation construction as in [FFF]. His finitely presented torsion-free property-(T) group P contains every finitely presented torsion-free group, in particular a direct product $P_1 \times P_2$ with $P_i \cong P$. Choosing an isomorphism $\alpha\colon P \to P_1 \leq P$ and any $a \in P_2 \setminus \{1\}$ gives the input required by Theorem 5.1. The group $E(P, \alpha, a)$ has torsion even though P does not, since the construction adjoins the involution c.

S14. The maximal group C^*-algebra

For a countable group H, the maximal group C^*-algebra $C^*_{\max}(H)$ is the completion of the group ring $\mathbb{C}[H]$ in the norm

$$\|x\|_{\max} = \sup_{\pi} \|\pi(x)\|,$$

the supremum taken over unitary representations of H; the supremum is finite because $\|\pi(\sum_h \lambda_h h)\| \leq \sum_h |\lambda_h|$. Write u_h for the canonical unitary of $h \in H$.

Proposition S14.1 (the universal property)**.** *The canonical map $h \mapsto u_h$ is injective, and for every unital C^*-algebra B and every homomorphism $\rho\colon H \to \mathcal{U}(B)$, there is a unique unital $*$-homomorphism $C^*_{\max}(H) \to B$ with $u_h \mapsto \rho(h)$ for every $h \in H$.*

Injectivity follows from the left regular representation, and uniqueness follows from density of $\mathbb{C}[H]$.

Proposition S14.2 (a strictly dominated projection gives a proper isometry)**.** *Let A be a unital C^*-algebra containing a projection p and a unitary u with $p < upu^*$. Then A contains an isometry that is not a unitary; consequently A is not stably finite and has no faithful tracial state.*

Proof. Write $q = upu^*$, so q is a projection with $pq = qp = p$, and put $r = pu^*$. Then

$$r^*r = up\,pu^* = upu^* = q, \qquad rr^* = pu^*up = p.$$

Put $s = r + (1-q)$. Since $p(1-q) = 0$ and $(1-q)p = 0$,

$$s^*s = r^*r + (1-q) = q + (1-q) = 1,$$

and since $pu^*(1-q) = pu^* - pu^*upu^* = pu^* - pu^* = 0$ and likewise $(1-q)up = 0$,

$$ss^* = rr^* + (1-q) = p + (1-q) = 1 - (q-p).$$

Thus $s^*s = 1$ but $ss^* = 1-(q-p) \neq 1$, so s is a nonunitary isometry. A unital C^*-algebra containing a nonunitary isometry is not finite, hence not stably finite. If τ were a faithful tracial state then $\tau(1-ss^*) = \tau(1)-\tau(s^*s) = 0$ while $1 - ss^* = q - p$ is a nonzero positive element, contradicting faithfulness. □

Remark S14.3 (the maximal algebra under proper one-sided conjugation). Let G contain a property-(T) subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$; the pair arising from the base subgroup of E has this form. Let $P \in C^*_{\max}(G)$ be the spectral projection at 1 of the Hermitian Kazhdan average $\frac{1}{2|S|}\sum_{s\in S}(u_s + u_s^*)$, which exists by the spectral gap of Section 3. Then $P \leq u_t P u_t^*$, and the inequality is strict: in the quasi-regular representation on $\ell^2(G/t\Gamma t^{-1})$ the point mass at the base coset is fixed by $t\Gamma t^{-1}$ but moved by every element of $\Gamma \setminus t\Gamma t^{-1}$, so the two fixed-space projections are distinct in this representation. By Proposition S14.2, $C^*_{\max}(G)$ is then not stably finite, has no faithful trace, and in particular is neither residually finite-dimensional nor MF. The reduced algebra $C^*_{\mathrm{r}}(E)$ behaves differently: its canonical trace is faithful and it is stably finite, despite being non-MF (Theorem C).

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